

Beliefs and Actions under Government Policy Uncertainty: Evidence from Student Loan Forgiveness*

Dmitri Koustas, Michael Weber and Constantine Yannelis

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Abstract

How does uncertainty about future government policy affect households' beliefs and subsequent spending and debt payment behavior? We study these questions through the lens of student loan forgiveness in the United States, which following electoral promises, was announced in 2022 but never implemented due to judicial rulings. We conduct a customized information provision experiment embedded in a survey eliciting real-time beliefs about future debt forgiveness and repayment, which we link to credit bureau data, employment verification data, and nondurables consumption. Borrowers who are more optimistic about forgiveness reduce payments on student loans by \$40 per month and increase non-durable spending by \$80 per month. We also find evidence that optimistic borrowers may postpone durable spending while waiting for uncertainty to resolve. Borrowers optimistic about future payment pauses make fewer payments on their student loans, reduce payments by \$40 per month and are 7.5 percentage points more likely to be delinquent after payments resume. A calibrated model of repayment choice implies that overly optimistic beliefs cost borrowers up to 6.9 percent of the initial loan balance. Our results provide micro-evidence on the role of policy uncertainty in household decision-making, and have implications for government announcements and commitment policy.

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1 Introduction

Governments often make fiscal promises to households, such as commitments to provide targeted benefits or cut taxes. These assurances frequently come with substantial uncertainty due to political, economic and legal wrangling across multiple branches of government. How do households' beliefs about future government policy affect their behavior today? And does uncertainty in the implementation of these policies affect how consumers respond?

In this paper, we study how individuals' expectations affect financial and economic decision-making under uncertainty about future fiscal transfers, exploring consumers' beliefs and their choices in response to promises of future benefits through a policy that underwent significant tumult after 2020: student loan forgiveness.¹ Student loan forgiveness was proposed several times by lawmakers in the U.S., but came to the forefront during the 2020 U.S. Presidential election. After different forgiveness plans were proposed during the election, President Biden announced in August 2022 forgiveness via executive order. This promised forgiveness was swiftly blocked by courts and followed by a year-long legal battle which ended in the reversal of broad loan cancellation. Exploiting the period of high uncertainty induced by the challenges of the policy in courts, we ran surveys eliciting expectations of anticipated student loan forgiveness along an extensive margin, whether or not it occurs, and also along an intensive margin, by how much, conditional on the implementation of the policy. We linked this survey to credit report and consumption data to study how beliefs about future government policy affect behavior.

We have three main findings. First, we document significant heterogeneity in beliefs about the likelihood of forgiveness and about the amount. Second, belief heterogeneity translates

¹In this paper, we are using a colloquial definition of uncertainty regarding whether government actions are known in advance. This aligns with the standard definition that uncertainty is a situation in which something is not known, or something is not known with certainty and is often used in the economics literature (Baker, Bloom, and Davis, 2016). At least two literatures in economics use separate and distinct different definitions of uncertainty. A large literature (Bloom, 2009; Luttmer and Samwick, 2018) uses uncertainty primarily to mean the second moment, while an older literature beginning with Knight (1965) uses uncertainty to refer to what economists all call ambiguity (Hansen, 2014; Barnett, Buchak, and Yannelis, 2023), or a situation where precise probabilities cannot be mapped to states of the world.

into behavioral responses. Individuals who have higher beliefs in debt forgiveness make lower monthly payments and are less likely to pay down their student loans. Finally, to study the aggregate implications of our findings at the micro level, we show that mistaken beliefs lead to significant welfare consequences for households.

We begin with a description of the survey instrument, and of how beliefs shifted with the policy environment. Our survey documents wide variation in beliefs about student loan forgiveness. For instance, in mid 2022, the mean belief about one-year ahead forgiveness was 25 percent, with a standard deviation of 30 percent. We show that these beliefs are sensitive to both news about policy events and experimental treatments regarding the likelihood of loan forgiveness. As part of our survey, we conducted randomized information interventions by presenting news articles that varied the degree of optimism about forgiveness. We repeatedly surveyed consumers over time, which allows us to study the response to major news shocks. Reading an optimistic news article increases forgiveness beliefs by 4 percentage points in our experimental setting and the Biden administration's major student loan announcement in August 2022 moved beliefs by 22 percentage points. Announcements about court injunctions and the Supreme Court decision in June 2023 saw beliefs fall back to their pre-announcement levels. These beliefs do not appear to evolve differentially due to pandemic risk or employment shocks faced during the COVID-19 period.

With our linked credit bureau data, we then document how households act on their beliefs about forgiveness. Correlating beliefs about student loan forgiveness with payment behavior, we show that those who are more optimistic about forgiveness/payment moratoriums make fewer payments on their student loans and increase non-durable spending. Individuals in the top two quartiles in terms of the likelihood of loan forgiveness are 30 percentage points less likely to make payments, and on average pay \$100 less per month on their student loans. Over the whole period, we see lower rates of durables spending among those who are more optimistic, even after conditioning on other observable differences, suggesting that those with more optimistic beliefs may delay large durable purchases until after the uncertainty

is resolved, consistent with real options theory with adjustment frictions (Bernanke, 1983).

These correlations, while suggestive, could be driven by unobservable differences among student borrowers with differing beliefs about the likelihood of student loan forgiveness. We address this concern in two ways. First, we utilize the timing and eligibility criteria of both policy announcements and legal reversals. Second, we exploit our randomized information treatments, which shifted individuals' beliefs about the chances of student loan forgiveness. Both strategies are consistent with changes in beliefs having a causal impact on loan repayments.

We further show that mistaken beliefs about forgiveness led to future adverse impacts for borrowers. Borrowers who were more optimistic that the pause would be extended made fewer payments once repayment restarted, reduced monthly payments by roughly \$40, and were subsequently more likely to become delinquent on their loans once payments restarted. In particular, among borrowers with payments due, those with above-median beliefs that the pause would be extended were 7.5 percentage points more likely to be 90 days past due in May 2025. These results show that mistaken beliefs about temporary relief policies can have persistent consequences: borrowers who expected continued forbearance adjusted their repayment behavior in ways that increased delinquency risk once uncertainty was resolved.

We also quantify the welfare consequences of mistaken beliefs about future forgiveness using a simple model of repayment choice and consumption. The model highlights that belief errors matter not only because they affect monthly payments, but also because they can lead borrowers to choose repayment paths that are ex post costly once uncertainty is resolved. Within a fixed repayment plan, welfare losses from incorrect beliefs are relatively modest. However, losses become substantially larger when beliefs affect repayment-term choices. If forgiveness ultimately occurs, pessimistic borrowers who repay too quickly can experience losses of 43.9 percent of the initial loan balance because they pay down balances that would otherwise have been forgiven. If forgiveness does not occur, optimistic borrowers who extend repayment in anticipation of relief incur smaller but still meaningful losses, up to 6.9 percent of

the initial loan balance, due to additional interest and delayed repayment. These results show that ambiguous policy signals can generate economically significant welfare costs by distorting household repayment and consumption decisions before policy uncertainty is resolved.

Our study relates to several strands of the literature. Most directly, we contribute to the growing literature on student loans. With \$1.6 trillion in student loans outstanding, they have become the second largest component of household balance sheets in the United States after mortgages. We are most closely related to the student loan literature looking at the effects of balances and payments (Boutros, Clara, and Gomes, 2022; Dinerstein, Yannelis, and Chen, 2024; de Silva, 2025; Katz, 2023; Black, Denning, Dettling, Goodman, and Turner, 2023; Briones, Powell, and Turner, 2023) and how it is related to non-education activities upon entering repayment (Avery and Turner, 2012; Looney and Yannelis, 2015; Lochner and Monge-Naranjo, 2011, 2016; Goodman, Isen, and Yannelis, 2021). Most directly, the paper is also related to a literature on student loan forgiveness (Di Maggio, Kalda, and Yao, 2019; Dinerstein, Earnest, Koustas, and Yannelis, 2024; Catherine and Yannelis, 2022; Salman and Xie, 2026).² Our paper contributes to this literature by showing that, in a policy area with substantial uncertainty, belief heterogeneity can affect agents' actions. This finding has important implications for policy guidance, as statements from policymakers can lead to changes in behavior that have significant welfare consequences for borrowers.

More broadly, we examine consumption and other consumer choices under uncertainty. Economists have noted since at least Ricardo (1821) that beliefs about future actions can affect present behavior. A large body of work exists studying policy uncertainty, for example Bloom (2009) and Baker, Bloom, and Davis (2016), but only recently has the literature been able to study how macroeconomic uncertainty causally affects household behavior (e.g. Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024).³ Relatively little empirical work

²Yannelis and Tracey (2022) and Looney and Yannelis (2024) provide thorough reviews of the student loan literature.

³There is a related literature on the response of firms to uncertainty. See, e.g. Bernanke (1983); Baker, Bloom, and Davis (2016); Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2018); Kumar, Gorodnichenko, and Coibion (2023).

studies how uncertainty about government policy can affect household decision making. In theory, expectations about government policy can affect individual actions (Barro, 1974) and lead to large welfare costs (Luttmer and Samwick, 2018). However, direct empirical evidence is relatively scant, likely due to the difficulty in linking measured beliefs to outcome data in periods of significant uncertainty. We contribute to this literature by showing direct evidence that shifts in beliefs affect real economic outcomes.⁴

Finally, we build on and add to the literature on subjective expectations. For decades, economists had largely ignored the subjective expectations of households given that the leading assumption of full information rational expectations directly implies the model-consistent belief of the agent. Given the large dispersion in the cross section (Mankiw, Reis, and Wolfers, 2003) and biases in time series, Prescott (1977) famously said “like utility, expectations are not observed, and surveys cannot be used.” Over the course of the following decades, this view has largely changed and methodological advances in survey design and methodology have contributed to this change (Manski, 2004). A recent body of work shows that the dispersion in expectations at any point in time as well as the apparent biases when evaluated against ex-post realizations can largely be explained by the signals consumers use to form their expectations. For example, in the domain of aggregate inflation expectations, D’Acunto, Malmendier, Ospina, and Weber (2021) and D’Acunto, Malmendier, and Weber (2021) show that consumers focus on the price changes of goods in their own consumption bundle in line with the Lucas (1972) island model, especially of goods they purchase frequently, and put more weight on price increases rather than equal-sized price cuts. Together, these features can rationalize the large

⁴While not explicitly focusing on uncertainty, our paper also relates to studies on the response of consumption and debt to government policies. Most directly, our paper relates to papers studying loan relief (Agarwal, Amromin, Ben-David, Chomsisengphet, Piskorski, and Seru, 2017; Kluender, Mahoney, Wong, and Yin, 2025), and other lending policies such as household leverage (Boutros, Clara, and Kartashova, 2025; Mian and Sufi, 2011), monetary policy (Di Maggio, Kermani, Keys, Piskorski, Ramcharan, Seru, and Yao, 2017), technology adoption (Higgins, 2024) and quantitative easing (Di Maggio, Kermani, and Palmer, 2020). Several studies have also explored how debt and consumption respond to income and wealth shocks (Agarwal, Liu, and Souleles, 2007; Boutros, 2023; Mian and Sufi, 2009; Agarwal and Qian, 2014; Baker, 2018; Aydin, 2022), with some work explicitly focusing directly on uncertainty and beliefs (Indarte, Kluender, Malmendier, and Stepner, 2025; Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber, 2024; Antill, Jenke, and Kluender, 2023; Fermann, Kuhnen, Li, and Ben-David, 2024; Binder, 2017; Manski, 2004; Berwart, Higgins, Kulkarni, and Truffa, 2025).

dispersion but also the upward bias in expectations. But socioeconomic status (Das, Kuhnen, and Nagel, 2020), cognition (D’acunto, Hoang, Paloviita, and Weber, 2023; D’Acunto, Hoang, Paloviita, and Weber, 2019, 2021), and lifetime experiences also play a role (Goldfayn-Frank and Wohlfart, 2020).

Despite their apparent biases, subjective expectations still matter in determining consumers’ economic choices such as their housing decisions (Armona, Fuster, and Zafar, 2019), their consumption and savings decisions (Coibion, Gorodnichenko, and Weber, 2022), or their portfolio choice (Beutel and Weber, 2023; Schnorpfeil, Weber, and Hackethal, 2024; Laudenbach, Weber, Weber, and Wohlfart, 2024). We build on this work in that we implement a randomized information provision experiment embedded in a customized survey to establish causality rather than purely relying on correlations between survey responses and economic decisions. Weber, D’Acunto, Gorodnichenko, and Coibion (2022); D’Acunto, Malmendier, and Weber (2023) and D’Acunto and Weber (2024) summarize this literature for inflation expectations, Haaland, Roth, and Wohlfart (2023); Fuster and Zafar (2023) summarize the literature on information provision experiments in economics and provide guidance on how to design them, and Stantcheva (2023) provides best practices on how to design and run a survey. Bachmann, Topa, and van der Klaauw (2022) provide a handbook treatment on the elicitation, patterns, and use of subjective expectations in economics.

The remainder of this paper is organized as follows: In the next section, we discuss the institutional environment of student loans and forgiveness in our period of study. In Section 3, we present our new survey of beliefs about student loan forgiveness and discuss the headline findings from the survey. Section 4 outlines the individual data linkages between our survey and consumption, credit bureau, and employment data. Section 5 presents our main analysis using these linked data, examining the evolution of outcomes by beliefs over time, around key announcements, and in response to our survey experiments. Section 6 outlines a simple model of payment choice and consumption, which we use to understand the welfare consequences of our findings. Section 7 concludes.

2 Background and Institutional Details

2.1 Beliefs About Fiscal Policy and Individual Actions

A large body of work exists studying both empirically and theoretically how beliefs shape the transmission of monetary and fiscal policy. D’Acunto and Weber (2024) provide a recent review of the literature on expectations and household behavior. However, there is relatively little work studying how beliefs about government actions affect financial decision making.

Agents who anticipate debt forgiveness will defer making payments, and avoid prepaying debt. Thus, governmental signals about future policies may impact both loan repayment and consumption today. To motivate our survey design and the empirical analysis, we now show the baseline mechanism through the lens of a simple two-period model.

Assume agents are born with an exogenously-borrowed amount B_0 on which they cannot default, which carries interest rate r , and receive an exogenous income stream y_t in each period. These agents choose consumption and assets to carry forward in period 1, in order to maximize a standard discounted lifetime utility function with separable and well-behaved per-period utility.

If forgiveness of any remaining balance on the loan inclusive of interest were to occur with certainty in period 2, the optimal period 1 payment choice under full information is 0, if allowed by the lender.⁵ The household would maximize their lifetime utility subject to their lifetime wealth as if they had no debt: $y_1 + \frac{y_2}{1+r}$. Depending on their preferences and income path, they may seek to borrow more in period 1 or save through a separate asset, if available, to optimally smooth their consumption.

Now assume forgiveness is uncertain. In $t = 2$, with probability p any remaining debt including interest is forgiven, and with probability $1 - p$ the agent must pay back the remaining debt. Further assume $y_1 + \frac{y_2}{1+r} > B_0 > y_t$, that is, the initial balance is greater than any period

⁵Control over the timing of payments is ultimately a key distinction between the standard consumption and savings model and real-life debt repayment. We will later generalize this model in Section 6 to multiple periods, a separate borrowing/saving decision, and constrain repayments to an amortization schedule with a choice of payment term.

income but less than lifetime income, and that households have no access to a second asset to carry forward consumption tomorrow.⁶ Under these assumptions, the agent will optimally pay off some debt in period 1 to ensure against the risk of default if the loan is not forgiven in period 2. Let b_1^* be the optimal debt payment in period 1. The balance carried forward must be less than income in period 2, that is, $(B_0 - b_1^*)(1 + r) < y_2$, which serves as a “natural” borrowing constraint.

If forgiveness occurs, then $c_2^* = y_2$, and if not, then $c_2^* = y_2 - (B_0 - b_1^*)(1 + r)$. Hence, the Euler equation with uncertainty about loan forgiveness is given by:

$$u'(y_1 - b_1^*) = \beta(1 + r)[pu'(y_2) + (1 - p)u'(y_2 - (B_0 - b_1^*)(1 + r))]. \quad (1)$$

As the probability of forgiveness, p , approaches 0, the Euler equation becomes the standard Euler equation without forgiveness, and the household spreads payments over the two periods to optimally smooth their consumption. As p increases, more weight is placed on the higher consumption/lower marginal utility state, and the right-hand-side of the Euler equation decreases; in order for the Euler equation to hold, consumption in period 1 must rise, which is achieved by making a smaller payment today.

Note that because forgiveness is a binary event, p fully characterizes the household’s subjective distribution, and hence both its mean and its variance. To see the role of the second moment—what the literature typically terms “uncertainty”—we take a second-order Taylor expansion of u' around the mean of period-2 consumption, so that the right-hand side of (1) becomes:⁷

$$\mathbb{E}[u'(c_2)] \approx \underbrace{u'\left(y_2 - (1 - p)(B_0 - b_1)(1 + r)\right)}_{\text{(i) first moment}} + \underbrace{\frac{1}{2}u'''(\bar{c}_2)p(1 - p)(B_0 - b_1)^2(1 + r)^2}_{\text{(ii) second moment}} \quad (2)$$

⁶This assumption rules out self insurance via saving and waiting until the next period to pay off the loan after forgiveness is realized.

⁷Period-2 consumption equals y_2 with probability p and $y_2 - (B_0 - b_1)(1 + r)$ with probability $1 - p$, so the mean of period-2 consumption is $\bar{c}_2 = y_2 - (1 - p)(B_0 - b_1)(1 + r)$ and the variance is $p(1 - p)(B_0 - b_1)^2(1 + r)^2$. Taking a second-order Taylor expansion of u' around $\bar{c}_2$ (i.e. through the term in u''') and noting that the first-order term vanishes because $\mathbb{E}[c_2 - \bar{c}_2] = 0$ yields (2).

Term (i) says a higher p raises expected period-2 consumption, lowers expected marginal utility, and lowers the optimal period-1 payment. Term (ii) is a precautionary channel that works in the opposite direction, and is a second-moment effect. Under the common assumption of $u''' > 0$ it raises b_1^* and lowers c_1^* , since paying down the balance compresses the gap between the forgiveness and no-forgiveness states. Because term (ii) scales with $p(1 - p)$, it vanishes at $p = 0$ and $p = 1$ and is largest at $p = 1/2$, so that while b_1^* is always decreasing and c_1^* increasing in p , the relationship is non-linear.

We explore this intuition further through the context of the U.S. student loan market in the early 2020s, when student loan forgiveness was the subject of intense policy and legal debate.

2.2 Student Lending in the United States

Student loans play an important role in the funding of higher education in the United States, with around half of undergraduate and graduate students in recent years receiving at least some loans as part of aid packages (National Center for Education Statistics, 2023). As of 2020, nearly 43 million individuals held approximately \$1.6 trillion in outstanding student loans, which represents the largest source of household debt after mortgages (U.S. Department of Education, 2024; Board of Governors of the Federal Reserve System, 2024).

Student loans can be broadly categorized into federal and private loans, with over 92% of outstanding student loan debt being directly held or guaranteed by the federal government.⁸ Federal student loans come with several benefits, including fixed interest rates, income-driven repayment plans, and potential loan forgiveness options. Federal loans can be forgiven through Public Service Loan Forgiveness or Income-Driven Repayment plans, which tie student loan payments to a borrower's income and family size, extending repayment to 10–25 years before

⁸Guaranteed loans were issued under the Federal Family Education Loan Program (FFEL). Under FFEL, private lenders provided the loans, while the federal government guaranteed the loans against default and provided subsidies to lenders. The FFEL program was discontinued in 2010 and replaced with the Direct Loan Program, whereas extant FFEL loans continued to be serviced by loan servicers. Yet another type of loan is the Perkins loan, issued by participating schools, with funds provided by the federal government and the participating schools. The Perkins loan program was discontinued in 2017.

remaining balances can be forgiven. Historically, very few borrowers reached forgiveness due to strict requirements, but recent policy changes beginning in 2022—including payment count adjustments, the PSLF waiver, and the introduction of the SAVE plan—have dramatically expanded relief. Forgiveness resulting from these policy changes is studied in Dinerstein, Earnest, Koustas, and Yannelis (2024). The remaining student loan debt is held in private student loans, which also includes federal student loans that have since been refinanced with a private lender, typically at a lower interest rate. Importantly, private student loans are not eligible for any federal student loan relief. Whether federal or private, student loan debt is difficult to discharge in bankruptcy. Delinquency of 270 days or more will typically result in wage garnishment up to 15% of disposable income. In 2019, the year before a multi-year payment pause began, over 10% of student loan borrowers defaulted within 3 years (U.S. Department of Education, 2019).

2.3 Student Loan Forgiveness

As a result of rising student debt and the perceived burdens, calls for broad-based student loan forgiveness have increased in the years preceding the 2020 U.S. Presidential election. For most borrowers, student loan forgiveness was a remote possibility before 2020 but during that election the policy came to the forefront. Each major Democratic candidate promised significant student loan relief. The 2020 pandemic was a watershed moment for student loan payment relief. Almost immediately, student loan payments and interest were paused as part of a bipartisan pandemic relief package in early 2020, resulting in automatic rehabilitation of federal loans in default.⁹ The payment pause was then extended on eight separate occasions: Student loans were initially paused as part of the CARES Act, effective March 27, 2020, through September 2020.¹⁰ In the end, the initially temporary payment moratorium lasted

⁹FFEL loans did not qualify for the repayment pause (Dinerstein, Yannelis, and Chen, 2024).

¹⁰This was later extended (twice) through September 2021. In August 2021, the payment pause was extended through January 2022. In January 2022, the payment pause was extended through May 2022. In April 2022, the payment pause was extended through August 2022. In August 2022, the payment pause was extended to January 2023. The press release referred to this extension as the “final time” that this policy would be extended. Despite

3.5 years until October 2023.¹¹ The extensions after August 2022 were in part motivated by developments related to student loan forgiveness, which we discuss next.

In August 2022, the Biden administration proposed broad-based cancellation of student debt in line with a key campaign promise. Under the plan, borrowers would receive \$10,000 in loan forgiveness. Recipients of Pell grants, which are federal grants given to students from low-income households, could receive a higher amount of \$20,000. This proposed plan for forgiveness had an income threshold of \$125,000 per year for singles, and \$250,000 for married couples. Initially justified under legal authority granted to the federal government to forgive loans during national emergencies, the policy soon encountered political opposition and legal challenges. On November 11, 2022, court injunctions blocked new applications for forgiveness. Following this date, prospective borrowers seeking to apply for forgiveness visiting studentaid.gov saw a banner that read: *“Courts have issued orders blocking our student debt relief program. As a result, at this time, we are not accepting applications. We are seeking to overturn those orders. If you’ve already applied, we’ll hold your application.”* Ultimately, a Supreme Court decision on June 30, 2023 blocked the order, and no broad-based forgiveness was issued under this plan.¹²

During this time, the proposed policies and legal battles received significant media attention. We visualize this media attention in Figure 1. The figure shows the number of news articles in major news outlets related to student loan forgiveness, over the period from 2019 to 2024. We further classify those articles into “Optimistic,” “Neutral” and “Pessimistic” regarding the likelihood of forgiveness using ChatGPT 3.5 Turbo. The number of articles peaks

this, in November 2022, the payment pause was extended to June 2023. In June 2023, the payment pause was extended again through September 2023.

¹¹While interest was beginning to accrue again, the Biden administration also announced that there would be no other consequences for non-payment for the next year, including no reporting of a delinquency to credit bureaus. In October 2024, payments paused again for individuals who had enrolled in the new, more generous income-contingent repayment plans, known as SAVE plans. Enrolling in these plans would also later be blocked by the courts.

¹²There were two separate court cases filed to block loan forgiveness, which had alternative outcomes highlighting the significant uncertainty regarding outcomes during this time period. One case was unsuccessful due to reasons of standing, while the second case *Biden v. Nebraska* was successful in stopping implementation of loan discharge. The court argued that legislation, which was initially intended for those affected by war or national emergencies who held federal loans, did not allow the executive branch to discharge government loans.

around the August 2022 announcement related to broad-based student loan forgiveness. As the figure shows, significant heterogeneity in the media’s forecasts regarding the likelihood of loan forgiveness was present. Throughout the window studied in Figure 1, a mix of optimistic and pessimistic information exists. During this time additional political statements, forgiveness made under other legal authority, and the introduction of new income-driven repayment plans that can lead to forgiveness, have also occurred. Given the popular media and policy attention, many individuals likely changed their beliefs about forgiveness dramatically since 2020. We next discuss our novel survey and the link to spending, credit bureau and employment data.

3 Surveying Beliefs about Student Loan Forgiveness

3.1 Survey Design

The key goal of our survey was to elicit individual beliefs about student loan forgiveness and repayment that we could link to economic outcomes.¹³ We asked the following series of questions to borrowers to elicit beliefs: First, *“What is the percent chance AT LEAST SOME of your student loan debt will be forgiven in the next [1,2-5,10+] years?”*¹⁴ We then asked *“Which fraction of your student debt do you think will be forgiven in the next [1,2-5,10] years?”* We also explicitly asked, *“If student loan forgiveness is announced, what dollar value of your student loan debt do you think will be forgiven?”* Using a related question, we elicited expectations related to the payment moratorium: *“What is the percent chance the repayment pause will be extended again beyond [DATE LAST ANNOUNCED]?”* The survey also included a randomized information provision experiment showing optimistic and pessimistic news articles about forgiveness. The

¹³A copy of our main survey instrument is provided in Appendix D.

¹⁴A large literature elicits subjective probability distributions in order to measure uncertainty separately from point forecasts, typically by asking respondents to assign probabilities to bins over a continuous outcome such as inflation or income growth, as in the New York Fed’s Survey of Consumer Expectations. See Manski (2004) and Bachmann, Topa, and van der Klaauw (2022) for discussions of elicitation methods. Density elicitation is necessary in settings when a point forecast does not pin down the dispersion of the underlying distribution: two respondents reporting the same expected inflation rate may hold very different degrees of uncertainty about it. Because forgiveness is a binary event, the probability elicited here is sufficient to characterize a respondent’s subjective distribution, including its second moment.

goal of this treatment was to shift beliefs about the likelihood of forgiveness, which was successful as we show below. We report these information treatments in Appendix B.

We fielded our survey on two different samples, which we can, respectively, link to credit panel and consumption data. The first was a survey frame purchased through an anonymous credit bureau, while the second was obtained through a commercial survey company that tracks household spending. The credit panel sample frame consisted of 1.3 million emails of potential student loan borrowers. These emails were sourced by the credit bureau by looking at nine-digit zipcodes with high rates of student loan borrowers. Using this survey frame, we conducted a series of email surveys using Qualtrics. In May-June 2022, we piloted the survey, providing incentives via lottery prizes. We conducted our first main survey waves from June to August 2022, using approximately half of our non-Gmail sample frame.¹⁵ Prior to the Supreme Court decision, we reached out to the remaining sample beginning in April 2023 and we recontacted prior respondents. After the Supreme Court decision, we conducted an additional follow-up survey. Finally, we conducted one more follow-up survey in May 2025 to follow up on the evolution of beliefs after the 2024 election. In total, we received 8,922 completed responses from 4,802 unique respondents, which corresponds to a 0.9 percent response rate among non-Gmail email accounts.

Separately, we conducted our survey on The Numerator OmniPanel. Numerator is a company which specializes in generating high quality consumption data and consumer insights. The panel comprises consumers using the Receipt Hog app.¹⁶ We can link survey responses to high-frequency spending data derived from receipts scanned by consumers; we discuss the Numerator data in more detail in Section 4 below. We fielded one survey wave in July 2022 targeting 3,000 respondents who indicated in a screening question that they had current

¹⁵We began with 20,000 pilot responses, half of which were randomly incentivized with a chance to win one of three Amazon gift cards. We did not find that response rates were sensitive to providing this incentive. For this reason, we proceeded without survey incentives. We found substantially lower response rates among individuals with Google-based email addresses which we believe is due to the way the Google email server handles large volume messages being sent via Qualtrics. Approximately 60 percent of our sample frame emails were from a non-Gmail address.

¹⁶More information about the Numerator panel can be found at <https://www.numerator.com/omnipanel/>.

student loans. In the end, we received 3,028 completed responses in our Numerator sample.

3.2 Raw Survey Results

3.2.1 Beliefs about Forgiveness and Repayment

We now briefly describe key survey results. We report additional results from our survey in Appendix C. First, announcements about student loans are highly salient among our respondents.¹⁷ In terms of repayment activity, 91% of respondents reported they were aware of the repayment pause, and 71% reported not having made a student loan payment since March 2020. Additionally, 18% reported making a voluntary payment, while 10% reported not having eligible student loans and only 1% said “Didn’t know”. Following the August 24, 2022 forgiveness announcement, 88% reported awareness of the forgiveness plan. In our survey in Spring 2023, 86% reported being aware that the Supreme Court might block forgiveness.

We plot the time series of our one-year ahead forgiveness expectations across all our survey waves in Figure 2. Our first surveys in June 2022 show mean beliefs around 27%. Mean beliefs jump by approximately 20 percentage points (p.p.) at the August 24, 2022 announcement about broad-based forgiveness, but were still below 50%. In a post announcement follow-up question, we asked about sources of skepticism. Approximately half indicated skepticism about legality, while around one-third indicated uncertainty about individual eligibility, such as income and eligible loan types. By the time of the Supreme Court decision, average beliefs were around 5 p.p. below their pre-announcement mean, but still reasonably high at 22%.¹⁸ By May 2025, however, beliefs about forgiveness fell to around 9%.

We next turn to analyzing the distribution of these beliefs. We plot the full distribution of

¹⁷Salience of policy is typically associated with consumers being better informed about the policy environment and details, which suggests lower treatment effects in information provision experiments (Weber, Candia, Afrouzi, Ropele, Lluberas, Frache, Meyer, Kumar, Gorodnichenko, Georgarakos, et al., 2025), but at the same time can also result in stronger changes in economic decisions as compared to complex policies (D’Acunto, Hoang, and Weber, 2022).

¹⁸Indeed, some individuals ended up receiving forgiveness through Public Service Loan Forgiveness, Income Driven Repayment, and Borrower Defense Programs. See Dinerstein, Earnest, Koustas, and Yannelis (2024).

cross-sectional beliefs before and after the two major announcements in Figure 3.¹⁹ We see considerable variation in beliefs, with mass points at 0, 50 and 100 percent.²⁰ So far we have focused on one-year ahead beliefs. Appendix Figure A.4 reports the distributions at different forgiveness horizons. In general, forgiveness probabilities increase at longer horizons. The mass point around 50% falls, and the mass point near 100% nearly doubles moving from shorter horizons to 10-year ahead beliefs. Appendix Figure A.5 reports the distribution of the fraction of their student loans individuals expect to be forgiven across different horizons, restricted to those with positive beliefs about any forgiveness. The median fraction at the one-year horizon was 14%. Only 10% of respondents expect all their loans to be forgiven over the course of the next 5 years. Beliefs about receiving 100% forgiveness nearly double from 10 to 20% of respondents between the 2-5 and 10-year horizons.

In Figure 4, we show similar plots for beliefs about the repayment pause. Prior to the penultimate extension of the repayment pause, the average belief was a 50% chance that the payment pause would continue and only around 10% of respondents said 0%. Prior to the final extension of the pause, beliefs had fallen considerably and around 30% of respondents now had beliefs of 0%; however, many still believed another extension would occur. The average belief was a 30% chance of a further extension that, ex post, did not materialize.

3.2.2 Survey Intentions

In our surveys in 2023, we also asked about consumption intentions if respondents received forgiveness. In particular, our question asked households to consider: “Suppose you received

¹⁹The time series of the second moment of beliefs—average subjective “uncertainty” measured as mean $p(1 - p)$ —is shown in Appendix Figure A.2, and the time series of the cross-sectional standard deviation of beliefs—“policy disagreement”—in Appendix Figure A.3. The two measures move differently. Subjective uncertainty is roughly constant after the August 2022 announcement while disagreement jumps, and after the court injunctions the pattern reverses: uncertainty rises while disagreement falls, indicating that households converged on a common view while each becoming individually less confident. Subjective uncertainty falls after the Supreme Court decision while disagreement remains constant. Both measures fall sharply after the 2024 election.

²⁰This pattern is another example that consumers were uncertain about the ultimate policy given that previous research shows that mass points at round numbers are typically a reflection of uncertainty (Binder, 2017; D’acunto, Hoang, Paloviita, and Weber, 2023).

student loan forgiveness of \$10,000 in July 2023.”²¹ We then prompted households to indicate how much of the \$10,000 they would allocate to new spending, such as a house/apartment; a car or other vehicle; a large home appliance or electronics; or all other spending. We also asked about saving. To summarize the results, 14.7% reported they would allocate money to new spending on housing, 15.5% reported they would allocate money to new vehicle spending, 8.7% reported they would allocate money to a large appliance/electronics, and 19.5% reported increasing other spending. Only 28.5% reported they would save at least \$1.

To examine how these spending intentions vary with beliefs, in Appendix Table A.1, we run a regression of spending intentions on one-year ahead forgiveness beliefs. Optimists are more likely to say they would increase spending on housing upon receiving forgiveness. The point estimate is positive, although insignificant, for autos. No strong pattern arises for other durables or other spending. The pattern for housing holds even after controlling for self-reported savings balances,²² suggesting these results for durables are not driven by liquidity (Weidner, Kaplan, and Violante, 2014). While not causal, these correlations provide suggestive evidence that those with more optimistic beliefs may be holding back more on large durable purchases—home purchases in particular—until forgiveness is realized, a point we will return to in our analysis below.

4 Data Linkages

This section describes linkages between our survey and credit bureau, employment and spending data.

²¹For respondents who self-reported a balance less than \$10,000, they were asked, “Suppose your student loans are completely forgiven in July 2023.”

²²Our survey asked respondents, “Approximately how much savings do you have available in your bank accounts?”

4.1 Anonymous Credit Bureau

4.1.1 Data Linkage

We sent completed survey responses to our credit bureau partners via secure transfer.²³ The credit bureau then conducted linkages using email addresses as the identifier. The linked data were anonymized by the credit bureau before being made available to us for analytical purposes. Because the survey frame was also sourced from the credit bureau, the match rate is high (>90%). We also linked a random sample of 100,000 non-respondents from the survey frame.²⁴ We have access to credit bureau outcomes from these credit bureau records from 2013 to the present. These data include standard outcomes such as payments, balances, loan inquiries and credit score (VantageScore 4.0). Detailed demographic information on age, gender, and educational attainment is also available for >90% of our matched respondents, and around half of the credit bureau universe. For around one-third of our sample, we also have employment records. These data are collected for employment and income verification purposes, such as when applying for credit or seeking new employment.²⁵

Appendix Table A.2 compares descriptive statistics between our respondents and non-respondents in the sample frame. Some differences arise. Notably, survey respondents have substantially higher student loan balances. The average (median) balance of respondents is \$73,100 (\$50,800) compared to \$49,300 (\$29,400) among the full sample frame. One explanation for the higher balances among respondents may be that respondents have greater benefits of gathering information about forgiveness and as such participate in the survey (Maćkowiak, Matějka, and Wiederholt, 2023; Weber, Candia, Afrouzi, Ropele, Lluberas, Frache, Meyer, Kumar, Gorodnichenko, Georgarakos, et al., 2025). Differences in other

²³Due to legal limitations, including data sharing agreements, we cannot disclose the name of the specific credit bureau from which the consumer credit data was obtained for this analysis.

²⁴This sample includes the non-surveyed Gmail email addresses.

²⁵Similar data are used in Gopalan, Hamilton, Kalda, and Sovich (2021) and Derenoncourt and Weil (2025). These data provide national coverage of employment history and monthly earnings from over 3 million employers, including government and non-profits, with around 50 million active records per month, which is just over 30% of total non-farm employment from the BLS establishment survey. Conditional on working full-year, we find that annual earnings closely compare to the CPS ASEC. Further details on coverage and comparisons to other employment and earnings data can be found in Dinerstein, Earnest, Koustas, and Yannelis (2024).

outcomes are small in terms of their economic significance. Appendix Table [A.2](#) also reports differences in employment between respondents and the same frame. Respondents are slightly more likely to have an employment match (1.8 p.p.), but have similar monthly earnings conditional on earnings.

4.1.2 Sample Restrictions

We restrict the analysis to individuals who have student loans in the credit bureau data. One consequence of our sampling approach is that the household member with student loans may differ from the person whose email is included in our sample frame. For instance, suppose we have the email of a parent who does not have student loans and we emailed this parent. It is possible they forwarded the survey to their child who has student loans. We would then link to the parents' credit bureau file. Restricting to respondents who have student loans in the data effectively deals with this issue. We lose approximately 20 percent of the respondent sample as a result of this restriction.

4.2 Non-Durable Spending (Numerator)

Our consumption data are obtained from Numerator, which directly linked our survey to their data. These data therefore capture cash and debit transactions missing in credit bureau data. The time period of the data is from 2017 through June 2023.

5 How Households Act On Their Beliefs About Forgiveness

5.1 Descriptive Analysis

Our main focus is on student loan repayment activity. As discussed above, 18% of our survey respondents reported making voluntary payments. Figure [A.1](#) shows repayment activity over this period. While it is declining over time, around 5% of individuals are making payments in

any month without having a payment due.²⁶ Thus, student loan repayment decisions are likely an important choice borrowers can make in this period, and may be affected by beliefs about future forgiveness.

We begin by examining raw summary statistics at the time of the survey among our respondents in Table 1. We report summary statistics by above/below median one-year ahead forgiveness beliefs at the time of the first survey, which are 19%. Those with above- and below-median beliefs appear similar in terms of student loan balances. However, those with more optimistic beliefs are less likely to make payments and make smaller payments. Those who are more optimistic are on average one year older, but are no more or less likely to be female.

Some of the differences in the raw summary statistics at the time of the survey may reflect differences in the date of the survey (we saw above that beliefs fluctuate over time) as well as other shocks occurring during this period. We next partially control for these differences by running the following specification:

$$y_i = \sum_{q=2}^4 \beta_q \mathbf{1}\{\pi_i^1 \in \text{quartile } q\} + \delta'X_i + \epsilon_i. \quad (3)$$

In this specification, we discretize one-year ahead beliefs, π_i^1 , into quartiles and include the following vector of controls: month surveyed, state of residence, whether an individual made any student loan payments in 2019, had any mortgage in 2019, zipcode-level housing price changes between February 2020 and survey date, and county-level COVID cumulative mortality rates at survey date.

Figure 5, Panel a, plots the β_q coefficients, along with 90 and 95% confidence intervals, visualizing the relationship between student loan repayment outcomes and one-year ahead forgiveness beliefs. We find student loan payment activity declines in beliefs about forgiveness, with households expecting forgiveness being less likely to make a payment. We cannot reject

²⁶Around 20% still have payments due, which is a combination of FFEL loans (Dinerstein, Yannelis, and Chen, 2024) and private loans. Even on these loans, borrowers may still have some scope to avoid payments, such as by filing for forbearance.

that the change in the share of borrowers making any student loan payment is the same for respondents in quartile 2 of one-year-ahead forgiveness beliefs (4-18 percent beliefs about one-year ahead forgiveness) and respondents in quartile 1 (0-3 percent) at the 5 percent significance level, although they do differ at the 10 percent level. Quartiles 3 and 4 (19 percent or more) are significantly different from quartile 1, but not from each other.

We next examine the dynamics of outcomes by beliefs. To do so, we divide the sample into above and below median beliefs, and examine how outcomes evolve over each quarter. As discussed above, beliefs and subjective uncertainty are tightly linked in our setting: because forgiveness is binary, the subjective variance $p(1-p)$ is a function of p alone. This bears on our above- and below-median comparisons: since the median is 19 percent, the below-median group is homogeneous in having low subjective uncertainty while the above-median group contains most respondents with beliefs near one half (mean $p(1-p)$ of 0.04 versus 0.19). The optimist indicator thus captures both a higher expected transfer, which lowers payments, and greater subjective uncertainty, which raises them, so our estimates reflect the combined effect of the two channels rather than the response to expected relief alone.

In particular, we estimate the following specification:

$$y_{i,t} = \sum_t \beta_t \mathbf{1}\{Optimist_i\} \times \mathbf{1}\{t \in yq\} + \delta' X_{i,t} + \text{group trends} + \epsilon_{i,t}, \quad (4)$$

where $Optimist_i$ is defined as $[\pi_i^1 \geq \text{quartile 3}]$. We omit Q1 2020, so that results are relative to the quarter prior to the payment pause. The regression also includes controls for state time trends, survey wave time trends, in repayment in 2019 time trends, any mortgage in 2019 time trends, zipcode-level housing prices, and county-level cumulative COVID mortality.²⁷ Standard errors are clustered at the individual level.

Figure 5, Panel b, plots the β_t coefficients, along with 90 and 95% confidence intervals. Individuals who eventually have above-median beliefs start reducing payments on student

²⁷September 2022 is recoded in 2022:Q4 as the full month occurs post the August 24, 2022 forgiveness announcement.

loans in the quarters after the payment pause, and this effect builds up over time, with a rapid increase in late 2020 and early 2021, coinciding with the November 2020 Presidential election. During this electoral contest, student loan forgiveness was a significant electoral promise of the Biden campaign. Prior to the payment pause, no differential trend arose in repayment between the two groups. Much of the level difference shown in panel (a) is accounted for in panel (b) in the period since Q1 2020. This pattern is consistent with news data seen in Figure 1. Immediately following the payment pause and the 2020 presidential election is a period when news about forgiveness begins to spread. It is thus likely that household beliefs we measure later are formed in the quarters after the payment pause.

We report estimates for a number of additional credit panel outcomes in Appendix Figures A.11 and A.12. Most of the outcomes show no clear differences between those with above versus below median beliefs throughout the window, with a few exceptions. One is mortgage balances, which widen precisely in the period around Biden’s first announcement (Figure A.11(e)-(f)). Combined with our survey evidence on durables purchase intentions, one possible explanation is that those with above-median beliefs are holding off making new home purchases while they wait for uncertainty to resolve. Revolving balances (i.e. credit card balances) also increase less for optimists (Figure A.11(b)), however, we do not see differences in total spending in Numerator (Figures A.12). This pattern is potentially consistent with optimists shifting funds originally earmarked to student loans to paying down non-student loan balances if they believe that their student loans are going to be forgiven. We see little effect on employment outcomes or auto loan balances.

5.2 Reactions Around Policy Announcements

The previous analysis shows point-in-time forgiveness beliefs are correlated with repayment activity, and these differences appear beginning in 2020 as forgiveness became part of national discourse. We next focus on *changes* in beliefs, by examining the period around major announcements. To do so, we run event studies around the three key events that occur during

our sample: (1) the initial forgiveness executive order made on August 24, 2022; (2) the Supreme Court decision blocking forgiveness made on June 30, 2023; and (3) the resumption of payments in October 2023. We stratify by prior beliefs about forgiveness, comparing those with above and below median beliefs. Predictions and estimation differ slightly for each event. We discuss each event and the results in turn.

5.2.1 Forgiveness Announcement

On August 24, 2022, President Biden, through executive action, announced \$10,000 in forgiveness for most borrowers. This announcement is therefore primarily a shock to those who were pessimistic about forgiveness prior to the announcement. We have two key predictions relating to debt prepayment and spending. First, pessimists with less than \$10,000 in balances should no longer want to make voluntary payments, while those with higher balances may still want to make payments as balances in excess of \$10,000 were not subject to the order. Second, pessimists will increase spending in response to an increase in expected wealth.

To test these hypotheses, we consider the following event-study specification:

$$y_{i,t} = \sum \beta_k \mathbb{1}\{t \in k\} \times \mathbb{1}\{Pessimist^{pre}\} + \alpha_i + \alpha_t + \epsilon_{i,t}, \quad (5)$$

where $Pessimist^{pre}$ is an indicator for below median beliefs, defined in surveys prior to the announcement. We run event study regressions separately for respondents with pre-announcement balances of \$10,000 or less (and hence who would have all their loans forgiven under the announced plan), and those with \$10,000 or more.

We plot the β_k coefficients tracing out the time-path of spending for pessimists, relative to optimists, in the top row of Figure 6. Panel (a) reports the coefficients for pessimists with balances under \$10,000. Panel (b) reports the coefficients for pessimists with larger balances relative to optimists with larger balances. Results from a more parsimonious specification pooling the event study coefficients into a “post” period and interacting with an indicator for

balances under \$10,000 are reported in Table 2. The first column in Table 2 reports the change in beliefs among the subset of respondents with a second belief observation post announcement in the event window.²⁸ On average, beliefs about one-year ahead forgiveness increase 20 p.p. among pessimists compared to optimists. No statistically significant difference in the change in beliefs among pessimists with balances below \$10,000 is present. Columns (2)-(3) report changes in student loan repayment activity. We find that pessimists with balances under \$10,000 reduce the probability of making a payment by about 7.7 percentage points and make \$40 lower payments in the post period, on average. As also evident in Figure 6 panel (b), no significant response among pessimists with balances above \$10,000 is detectable.

Figure 7 and Table 3 report results on spending in the Numerator data. We find that pessimists with lower balances increase non-durable spending by approximately \$80 per month. While this point estimate is greater than the monthly cash flow savings of approximately \$38, we cannot reject equality. The spending results are driven by non-food expenditure, suggesting that the increase in spending may come from more discretionary spending categories.

We report additional results in Table A.3a. In contrast to Numerator consumption, which largely captures non-durable and semi-durable spending, pessimists with low balances appear to *reduce* balances on auto loans, with about \$2,600 lower balances in the post period. One explanation could be that these pessimists update their beliefs and are now using savings from student loan payments to pay down auto debts.

5.2.2 Supreme Court Decision

On June 30, 2023, following legal challenges, the Supreme Court blocked the broad-based forgiveness plan. This event is thus a shock to those who were optimistic about forgiveness in the period prior to the announcement. We consider a similar event-study framework, this time

²⁸We do not run a regression on changes in beliefs given we only have post-period beliefs for about 10% of respondents.

comparing the response of optimists relative to pessimists:

$$y_{i,t} = \sum \beta_k \mathbb{1}\{t \in k\} \times \mathbb{1}\{Optimist^{pre}\} + \alpha_i + \alpha_t + \epsilon_{i,t}, \quad (6)$$

where $Optimist^{pre}$ is now an indicator for above-median beliefs, and is defined in surveys post announcement but prior to the Supreme Court decision. For this analysis, we restrict the sample to those who were surveyed after the injunctions in November 2022 and prior to the Supreme Court decision.

We plot event study coefficients in the bottom row of Figure 6, and results from a more parsimonious specification pooling the event study coefficients are shown in Columns (4)-(6) of Table 2. Column (4) shows beliefs of optimists fell by 27 percentage points, which is comparable in magnitude to the size of the shock around the announcement. The dynamics shown in the figure are noisier, but the point estimate for student loan payments for those with balances under \$10,000 is positive and the change in probability of making any payment is comparable to what we saw for the forgiveness announcement. Figure 6(c) shows the largest point estimate is the last month in the event window, October 2023, which is when interest again begins to accrue.

5.2.3 Resumption of Payments Due and the On-Ramp to Repayment

The final events we consider are the resumption of student loan payments in October 2023 after 3.5 years of the payment pause, and the end of the “on-ramp to repayment” one year later. Appendix Figure A.1 shows the share making payments more than doubled around the end of the repayment pause, from around 20 percent to 50 percent of student loan borrowers. We are now also able to examine an additional important outcome when it comes to student loans: student loan delinquency. As discussed above, the payment pause and related forbearance policies meant that very few borrowers experienced delinquency after March 2020. After the payment pause ended in October 2023, interest began to accrue again; however, missed

payments were not reported to credit bureaus for one more year as part of the Department of Education’s on-ramp to repayment. As a result, non-payment beginning in October 2023 would only be reflected in credit reports starting in October 2024, implying that 90-day delinquencies would begin to appear in early 2025.

The resumption of repayments is a shock to optimists who believed that the payment pause would be extended further. We directly elicited beliefs about the repayment pause being extended, and we examine outcomes around the resumption of payments, stratified by these beliefs, by estimating the following specification:

$$y_{i,t} = \sum \beta_k \mathbb{1}\{t \in k\} \times \mathbb{1}\{Optimist^{Pre}\} + \alpha_i + \alpha_t + \epsilon_{i,t}, \quad (7)$$

where $\mathbb{1}\{Optimist^{Pre}\}$ is an indicator for above median beliefs about the repayment pause being extended beyond October 2023. For this analysis, we restrict our sample to those who were surveyed in July 2023 post-announcement, and prior to the repayment pause ending. When examining delinquency outcomes, we further restrict to borrowers with a non-zero payment due.²⁹

We plot the key event study coefficients in Figure 8. Those who are most optimistic about another extension of the repayment pause are less likely to make payments (and hence accrue more interest) after the resumption of payments. Further, this optimism is predictive of delinquency in 2025, with optimists being 7.5 percentage points more likely to be 90 days past due in May 2025. This effect fades away over time, presumably as optimists finally update their beliefs about the extension after being reported delinquent. We report results from a more parsimonious specification pooling selected post periods in Table 4.³⁰ The effects on payments and delinquency load on optimism about another extension of the payment pause, rather than

²⁹Because this status can vary over the estimation window, the estimation sample is no longer strictly balanced. We prefer this definition since it better maps to public statistics on delinquency rates. Results focusing on a balanced sample of individuals with balance due in December 2024 are qualitatively similar, after scaling by the share with a balance due.

³⁰For student loan payments, the table pools together 5 post periods to be consistent with earlier results. For delinquency, we pool together 2025:Q2, to account for lags in reporting observed in the event study.

on beliefs about future forgiveness.³¹

5.3 Experimental Variation

A potential concern is that beliefs are correlated with unobserved factors, which either also change during policy announcements, or which are heterogeneously impacted by policy announcements. To address this concern, we experimentally vary beliefs about the likelihood of loan forgiveness. As part of our survey, we conducted a series of information treatments. Over the course of our surveying, we presented three sets of optimistic and pessimistic articles: pre-8/24 Announcement (June-Aug 2022), post injunction (Nov 2022), and pre-Supreme Court decision (April-June 2023).

The articles used as treatments come from major news sources such as *The Wall Street Journal*, *CNN* and *USA Today*, and discuss either the determination of the Biden administration to press ahead with loan forgiveness, or various legal challenges that loan forgiveness would face. The text of sample treatments are shown in Appendix B. Each survey wave randomly assigned respondents to receive an article or to the control group with no article, with 1/3 probability. Some respondents who were resurveyed prior to the Supreme Court decision were treated twice, with both treatments being independent. In Appendix Table B.4, we report a balance table comparing observables between our two treatments and control groups, for the first treatment. The p-values on the F-statistic for the optimistic and pessimistic treatments are 0.942 and 0.567 respectively, indicating that the randomization was successful.

We report a first-stage regression of beliefs on various functional forms of the treatment in Table 5. We find that the optimistic treatment increased beliefs by around 3.7 percentage points, but find no effect of the pessimistic treatments (Columns 1-2).³² As shown in Column

³¹We report additional outcomes in Appendix Table A.4. We see some evidence that optimists reduce their mortgage balances. This may be because they have more cash flow to make payments on their mortgage since they are making fewer student loan payments.

³²This asymmetry is potentially consistent with “motivated beliefs,” a theory from the behavioral psychology literature which documents a tendency to down-weight negative information but could also be consistent with ambiguity aversion, see Epstein and Schneider (2008). Our event study around the Supreme Court decision did show beliefs fell dramatically, so it could be that people respond differently to pessimistic information that is more

(3), we cannot reject that the three different optimistic treatments we used at different times had the same impact on beliefs. Results are similar restricting to those we link to positive student loan balances in the credit bureau data and who report having no payment required (Column 4). These respondents will make up our IV main estimation sample for second stage outcomes. The F-statistic using the optimistic treatment is 15.45 in this case, above conventional rule of thumb thresholds for a weak instrument. There is no differential response for optimists with balances above or below \$10,000 (Column 5).

We next examine the effect of our optimistic treatment on outcomes by pooling the pessimistic treatment with the control group, and running separate rolling IV regressions around the survey date:

$$y_{it}^h = \beta_{IV}^h \widehat{\pi}^1 + \delta^h X_i + \epsilon_{it}^h$$

$$\pi_{it}^1 = b_1 \mathbb{1}\{Optimistic\} + d'X_i + u_{it},$$

where h denotes the time in relation to the survey date; $h = 0$ is the date of the survey, and $h < 0$ are placebo regressions prior to the date of the survey. $\mathbb{1}\{Optimistic\}$ is an indicator of having an optimistic treatment regarding loan forgiveness.

We plot the β_{IV} coefficients of making any student loan payment in Figure 9. Reassuringly, the placebo periods prior to the survey all show zero effects, which are both statistically indistinguishable from zero and very close to 0 in terms of magnitudes. Following our information intervention providing the optimistic treatment, student loan payments begin falling in the treatment group. Figure 9 shows a statistically significant reduction in payments following the treatment.

uncertain.

To maximize statistical power, we pool the post outcomes in an IV event-study framework:

$$y_{it} = \beta_{IV} \pi^h \times \widehat{\mathbb{1}\{Post\}} + \beta_2 \mathbb{1}\{Post\} + \alpha_i + \alpha_t + \epsilon_{it}$$

$$\pi_{it}^h \times \mathbb{1}\{Post\} = b_1 \mathbb{1}\{Optimistic\} \times \mathbb{1}\{Post\} + b_2 \mathbb{1}\{Post\} + a_i + a_t + u_{it}, \quad (8)$$

where $\mathbb{1}\{Post\}$ is a post survey indicator. We exclude the treatment month due to partial treatment, and pool 4 periods pre and post to increase power. Standard errors are clustered at the individual level. We report results in Table 6. We find that a 10 p.p. increase in beliefs reduced the payment propensity by 3.72 p.p. Results on monthly payments are negative, consistent with our earlier results, but less precisely estimated.³³

Our results also provide some evidence that projected loan forgiveness led to changes in the intention to vote. In the first wave of our survey, we also asked about intentions to vote in the November 2022 midterm elections. We report results from the IV regression in Column (3) of Table 6. Interestingly, increasing forgiveness beliefs via our optimistic treatments increased voting intentions: A 10 p.p. increase in forgiveness beliefs increases propensity to vote by 11.4 p.p. Among our respondents, 58 percent reported intending to vote for the Democratic party.

6 Welfare Quantification

In this section, we extend the simple stylized model presented in Section 2 to illustrate how beliefs affect repayment choice and how corresponding uncertainty impacts borrower welfare. We use the model to quantify the welfare consequences of inaccurate beliefs, and show how distorted government signals about future policy actions can lead to substantial welfare losses.

³³Results for other credit bureau and employment outcomes are reported in Table A.5. However, results for these outcomes are not precisely estimated.

6.1 Model Setup

We assume a household starts life with student loan debt B_0 , subject to interest rate r_b . There is uncertainty about potential debt forgiveness of amount F , which may occur at a future period $T_F \leq T$, where T is the terminal period. We assume that the household can afford to repay the loan based on lifetime income, and that per-period income exceeds the amortized payment:

$$y_t > B_0 \frac{r_b(1+r_b)^\tau}{(1+r_b)^\tau - 1}, \quad \forall t \leq \tau. \quad (9)$$

where $\tau \leq T$ is the loan term. Households can also borrow and save at an interest rate r_s .

Households choose consumption $\mathbf{c} = \{c_1, c_2, \dots, c_T\}$ and a repayment plan $\mathbf{b} = \{b_1, b_2, \dots, b_\tau\}$ determined by the term choice τ , to maximize expected lifetime utility:³⁴

$$\max_{c_t, \tau} \sum_{t=1}^T \beta^{t-1} \mathbb{E}_t[u(c_t)], \quad (10)$$

where u is a twice-differentiable concave utility function and β is a discount factor. The household faces the lifetime budget constraint:

$$\sum_{t=1}^T (1+r_s)^{T-t} (y_t - b_t) = \sum_{t=1}^T (1+r_s)^{T-t} c_t. \quad (11)$$

Loan payments b_t depend on potential forgiveness F , which occurs at time T_F . If forgiveness does not occur, households continue to pay b_t until τ . If households receive forgiveness, payments are made until the remaining principal is eliminated (which may be the current period if the full balance is forgiven), after which no further payments are made.

The choice of repayment term, τ , is a central margin in the model. This choice governs how beliefs translate into behavior. With $r_b > r_s$, a shorter term minimizes total interest; a longer term minimizes payments made before relief might arrive, preserving a larger

³⁴The model assumes agents maximize subjective expected utility. Behavioral biases (e.g. debt aversion, default biases), may amplify or dampen existing forces in our model.

balance to be forgiven. Beliefs about forgiveness therefore determine which consideration dominates. We assume households may borrow and save freely at r_s , but repayments must follow the amortization schedule implied by the chosen term (no prepayments). We adopt the schedule-based restriction for tractability. An alternative would be to permit prepayments but impose a binding borrowing restriction for r_s , which would likewise prevent households from retiring the balance early.³⁵

Denote the discounted value of all repayments as $B := \sum_{t=1}^T (1 + r_s)^{T-t} b_t$. It then follows from the Euler equations and budget constraint that:

$$\frac{dc_t^*}{dB} < 0, \quad \forall t. \quad (12)$$

Hence, the household is better off under a repayment plan with lower B . The optimal consumption decision comes down to the optimal term choice, τ . When choosing the optimal repayment plan, the household takes into account both the interest rate on the loan and their expectation of forgiveness.

Intuitively, optimality requires increasing consumption before T_F in anticipation of possible forgiveness, reflecting the higher expected resources. The household's optimal repayment term τ balances the trade-off between minimizing interest costs and deferring payments in anticipation of forgiveness. If forgiveness is expected, longer repayment terms allow households to reduce payments before the debt relief, whereas if forgiveness is not expected, shorter terms minimize total interest.

³⁵This approach would also increase the contribution of the second-moment channel. When households can save freely and are unconstrained in borrowing, a shock to the remaining balance can be spread across the remaining horizon; the resulting consumption gap across states is an annuitized fraction of the balance rather than the balance itself, and since the variance term in (2) enters this gap squared, the precautionary channel is quantitatively second order. This contrasts with the two-period model of Section 2, which rules out a second asset and therefore maximizes the scope for the precautionary channel, since repayment is the only available hedge. A binding borrowing constraint would therefore raise the risk-bearing cost of misperceptions, both because the consumption gap across states would widen and because effective prudence rises near a binding constraint. Thus, our welfare estimates are a lower bound.

6.2 Model solution and calibration

We calibrate a numerical solution to our stylized model under various beliefs. We set the following parameters for our baseline calibration: a 25-year planning horizon ($T = 25$), CRRA utility with coefficient of risk aversion parameter $\rho = 2$, annual discount factor $\beta = 0.96$, and constant after-tax income of \$56,329 per year (mean in the employment data). The initial loan amount is $B_0 = \$50,000$, which is the median among our respondents. In our baseline calibration, we set r_s to 5 percent and r_b to 6.5 percent. This borrowing rate results in an amortized annual payment of \$6,955 if there were no payment pause.

We allow households to adjust to the shorter term once forgiveness is announced, so the welfare losses we report are those that re-optimization cannot undo: principal already retired on a balance that would otherwise have been forgiven.

To reflect other policy features, we assume a three-year interest-free payment pause at the start, followed by re-amortization over the remaining term. While in reality the end of the payment pause was not directly known, we assume no uncertainty in the end date of the repayment pause. We consider various forgiveness horizons which we discuss in the next section.

As is standard for consumption models under uncertainty, we solve the model using numerical methods. We solve the Euler equation at T_{F-1} to account for the expected marginal utility of post-forgiveness consumption. Consumption pre- and post- T_{F-1} is otherwise deterministic. In the post period, it is a function of assets held at T_{F-1} and the future stream of payments, which varies based on whether forgiveness was realized at T_F . Consumption prior to T_{F-1} is chosen to satisfy the intertemporal budget constraint. This approach produces a full consumption and asset path.

6.3 Model Results

Table 7 reports the willingness to pay for households with ex-post correct beliefs about debt forgiveness and those holding various ex-ante beliefs. Each cell shows the present value of the lifetime consumption a borrower would be willing to give up to be indifferent between the ex-post correct belief and their actual belief, expressed as a percent of the initial loan balance for ease of interpretation.³⁶ Columns (1)-(3) report consumption differences within a fixed repayment plan, while columns (4)-(5) report differences relative to the optimal repayment plan chosen under ex-post correct beliefs.

Several patterns emerge. Within a fixed repayment plan (columns 1–3), welfare losses from misperceived forgiveness are generally small. The most optimistic households that incorrectly anticipate forgiveness experience modest reductions in lifetime utility if forgiveness does not occur (column 1) with losses equal to at most 1.38 percent of the initial loan balance. The most pessimistic households that did not anticipate forgiveness experience slightly higher losses if forgiveness occurs (up to 1.74 percent). Note that there is no relative welfare loss if forgiveness occurs after 10 years under the 10-year repayment term. This is because the loan is paid off prior to the forgiveness realization for both optimists and pessimists.

Welfare differences are substantially larger when households choose repayment terms based on incorrect beliefs (columns 4–5). If forgiveness does occur, it is optimal to choose the 20-year repayment term, allowing households to minimize payments prior to debt relief. Pessimists who select the 10-year term experience large welfare losses of 43.85 percent of the initial loan balance, reflecting the fact that they paid down their loan prior to forgiveness. Conversely, if forgiveness does not occur, it is optimal under the calibration to select the 10-year term to minimize interest costs, and optimists who expected forgiveness and select the 20-year term incur smaller but still noticeable losses (up to 6.88 percent of the value

³⁶Appendix Table A.6 reports the equivalent variation, defined as the constant percent of consumption a borrower would need to give up under ex-post correct beliefs, to make utility equivalent to indicated beliefs. The values in Table 7 are calculated by multiplying the equivalent variation by the reference consumption path, discounting these values to present using the savings rate (r_s), and summing across all periods.

of the initial loan). These results highlight the interaction between beliefs and term choice: misperceptions about forgiveness timing or likelihood can lead to economically meaningful welfare losses, particularly when households choose repayment terms that are not aligned with ex-post optimal behavior. Belief errors are costly here not because households bear risk, but because ambiguous signals lead them to commit to repayment paths whose costs are largely sunk by the time uncertainty resolves.³⁷

Figure 10 considers the model results for the full set of beliefs and forgiveness realization horizons between years 4-10, with an endogenous term choice. Panel (a) shows the welfare losses if forgiveness does not occur, and panel (b) shows the welfare losses if forgiveness occurs.

The figure makes clear that misperceptions matter more as the expected forgiveness horizon extends. If forgiveness were to occur early—e.g. after the end of the 3-year payment pause—there is no welfare loss from choosing the wrong payment term choice since it is optimal to make no payments during the payment pause. Welfare losses are still present from consuming too much or too little, and these welfare losses from misperceptions are increasing in the forgiveness horizon. As the expected horizon to forgiveness gets longer, borrowers expecting forgiveness will want to reduce payments by extending the term length of the loan, even under modest forgiveness beliefs. Linking this back to our empirical findings on beliefs shown in A.4, we found that 65% of respondents had positive expectations of forgiveness 10 years in the future. Households anticipating forgiveness far into the future optimally increase early consumption and extend payments, paying more in interest up front. If forgiveness does not materialize, these early-period adjustments produce larger lifetime welfare losses. Upon realization, not receiving forgiveness generates welfare losses up to approximately 0.5% of lifetime consumption among optimists who were rationally choosing to make lower payments (Panel a). However, as shown in Panel b, the largest welfare losses occur among pessimists

³⁷The channel generating these losses differs from that in the literature on policy uncertainty. Luttmer and Samwick (2018) find substantial welfare costs from uncertainty over future Social Security rules, where the uncertainty is undiversifiable and households have no margin through which to reduce their exposure. In our setting the risk-bearing cost is small, as columns (1)–(3) indicate: incidence is largely idiosyncratic, repayment itself provides a hedge at a cost of only the spread between r_b and r_s , and resolution occurs early relative to the repayment term.

with incorrect beliefs as the expected forgiveness horizon grows. In order to avoid interest, pessimists will want to pay off their loans within 10 years. Pessimists with incorrect beliefs therefore experience the largest welfare losses relative to correct beliefs because they miss out on forgiveness, largely driven by the present value of the transfer.

Our modeling exercise, combined with the earlier empirical results, demonstrates that communication or actions leading to greater misperceptions can result in larger adjustments upon realization and greater welfare losses for borrowers. Further, these losses grow as the horizon under which uncertainty resolves extends further into the future.

7 Concluding Remarks

How households form and act on their beliefs is important for many economic models which economists have noted at least since Ricardo (1821). Moreover, policymakers often send mixed or inaccurate signals regarding future policy actions, and these signals can affect agents' beliefs. Misperceptions created by policy ambiguity can have significant welfare consequences and affect the aggregate economy. This paper provides new evidence regarding how government policy shapes beliefs and actions in the context of student loan forgiveness.

We provide three main sets of findings. First, we document substantial heterogeneity in borrowers' beliefs about student loan forgiveness and repayment policy. Even during periods when policy announcements were highly publicized, borrowers differed meaningfully in their expectations about whether forgiveness would occur, how much debt would be forgiven, and whether the payment pause would be extended. Beliefs responded strongly to major policy events, including the August 2022 forgiveness announcement and the subsequent legal reversals, but they did not converge to a common view. This dispersion suggests that households were not merely reacting mechanically to policy announcements; rather, they interpreted the policy environment through heterogeneous information, priors, and expectations.

Second, we show that these beliefs translated into economically meaningful behavior. Borrowers who were more optimistic about forgiveness made fewer student loan payments and paid less toward their balances. These effects appear both in observational patterns around policy events and in experimental evidence from randomized information provision, which shifted beliefs about forgiveness. The results indicate that expectations about future government action can causally affect present financial decisions. In addition, we find evidence that optimism about forgiveness was associated with higher non-durable spending and, in some cases, delayed durable purchases, consistent with the idea that households adjust both repayment and consumption decisions when the value of future debt obligations becomes uncertain.

Third, we document that mistaken beliefs can have important consequences when uncertainty is resolved. Borrowers who expected additional extensions of the repayment pause were less likely to resume payments and were more likely to become delinquent once repayment restarted. Our model further shows that welfare losses from incorrect beliefs can be economically significant, especially when beliefs affect repayment-term choices or when uncertainty persists for a long period. These losses arise not simply because borrowers are uninformed, but because ambiguous policy signals can make it rational for households to alter repayment and consumption decisions in ways that later prove costly once the policy outcome is known.

Our results have important implications for policymakers in terms of policy communications. Ambiguous or contradictory policy communications can lead to suboptimal consumer behavior, leading to significant welfare consequences. When policymakers announce policies that later face legal or political obstacles, households may adjust their spending, saving, borrowing, and repayment decisions in anticipation of benefits that may never arrive. This does not imply that governments should avoid communicating policy intentions. Rather, it highlights the value of clarity, credibility, and institutional feasibility in policy communication. Ambiguous or overly optimistic signals can induce households to make decisions that are

privately costly and potentially destabilizing when expectations are later reversed.

The student loan setting is especially interesting because borrowers had many options regarding repayment behavior during the payment pause, and policy debate about forgiveness was particularly salient. However, the mechanisms we document are likely to apply more broadly. While we study fiscal policy communication and beliefs in the context of loan forgiveness, similar effects may be present for taxes, pensions, tariffs, stimulus or other government fiscal policies. Studying these beliefs about future government fiscal policy, and whether there are similar or different effects, is potentially a fruitful avenue for future work.

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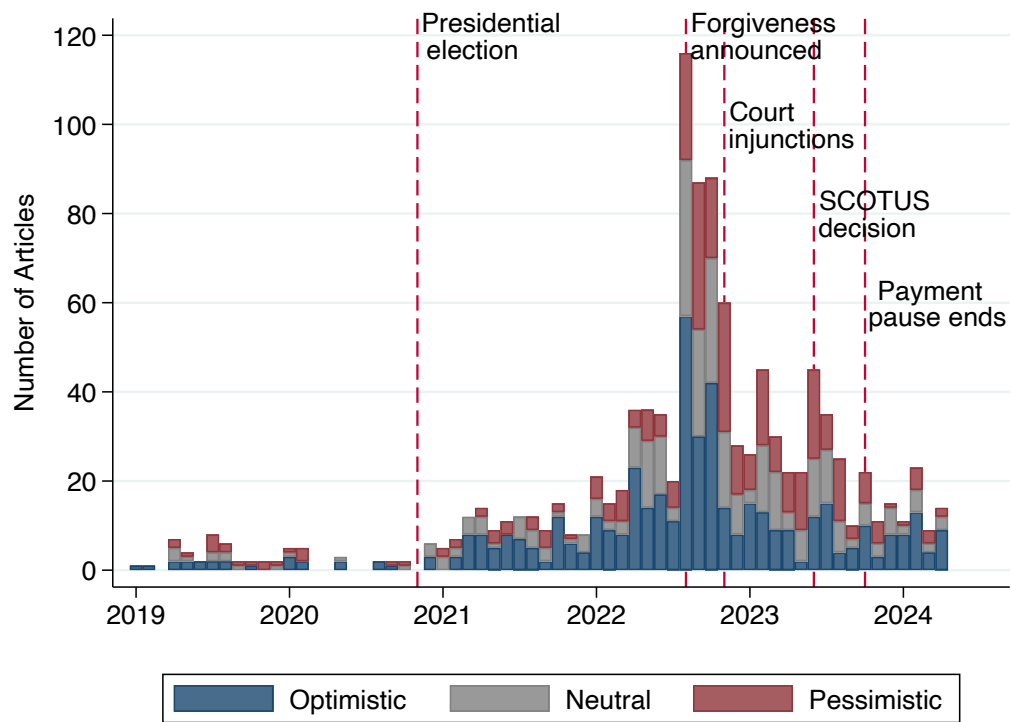
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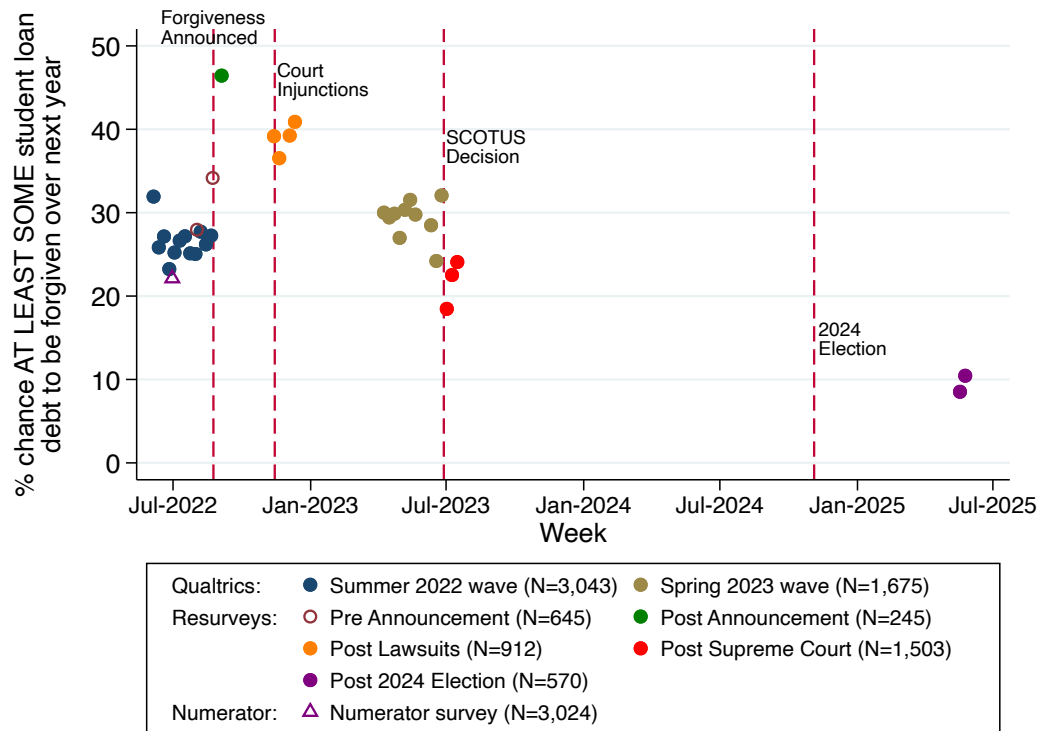
Figures

Figure 1: Key Dates Related to Student Loan Forgiveness and News Sentiment



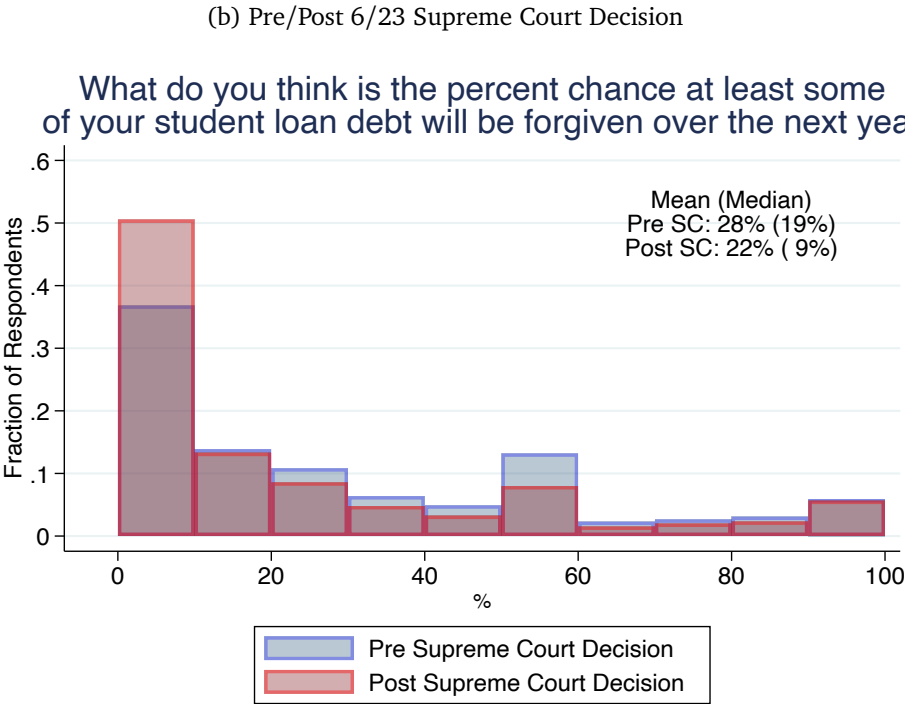
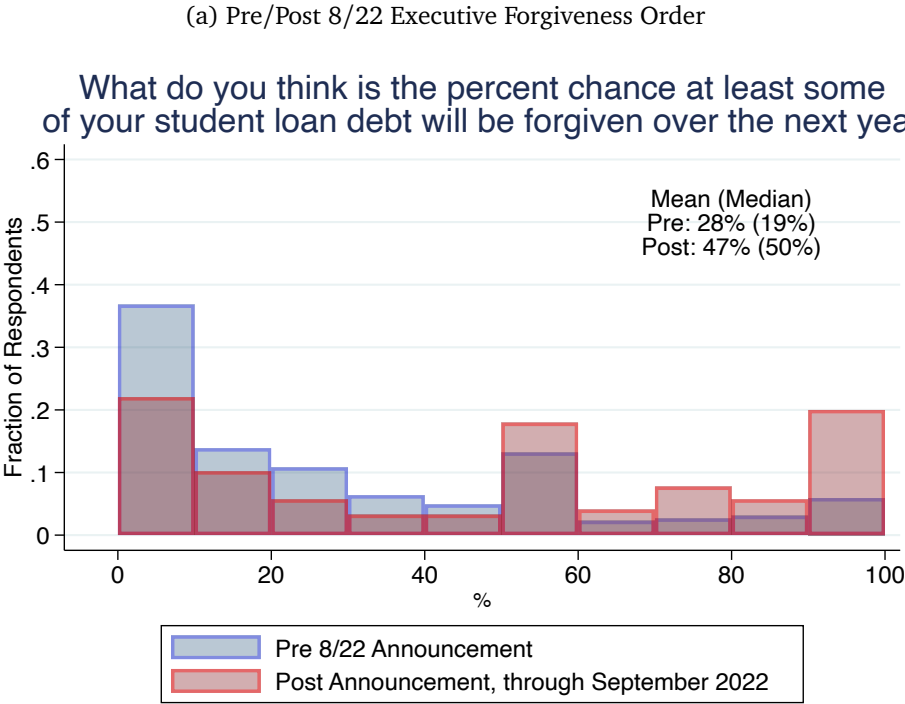
Notes: Figure shows the number and sentiment about news articles from major news sources (Business Insider, Los Angeles Times, New York Times, The Washington Post, USA Today, and Wall Street Journal). Articles sourced from UChicago ProQuest. Sentiment classified by Chat GPT 3.5 Turbo.

Figure 2: Survey Beliefs About 1-Year Ahead Forgiveness



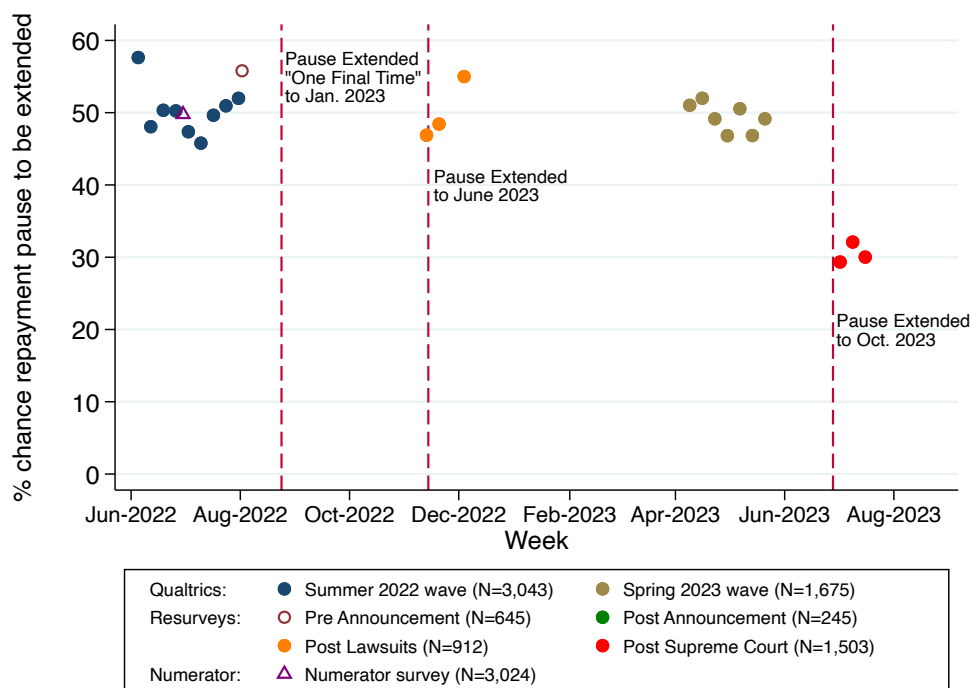
Notes: Figure shows mean beliefs about 1-year ahead forgiveness from our surveys. Each dot is an average over individuals surveyed in the week. Source: Qualtrics survey and Numerator survey.

Figure 3: Full Distribution of Beliefs, Pre and Post Major Policy Announcements

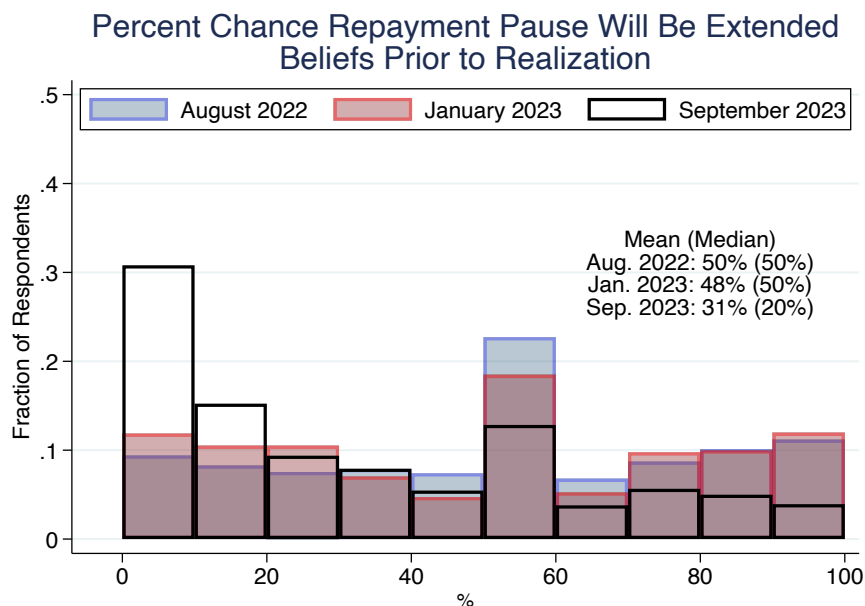


Notes: Figure shows the distribution of beliefs about 1-year ahead forgiveness. Panel (a) shows beliefs through September 2022. Panel (b) is restricted to beliefs collected after November 2022. Source: Qualtrics survey.

Figure 4: Expectations about the Resumption of Student Loan Payments
(a) Time-Series Means

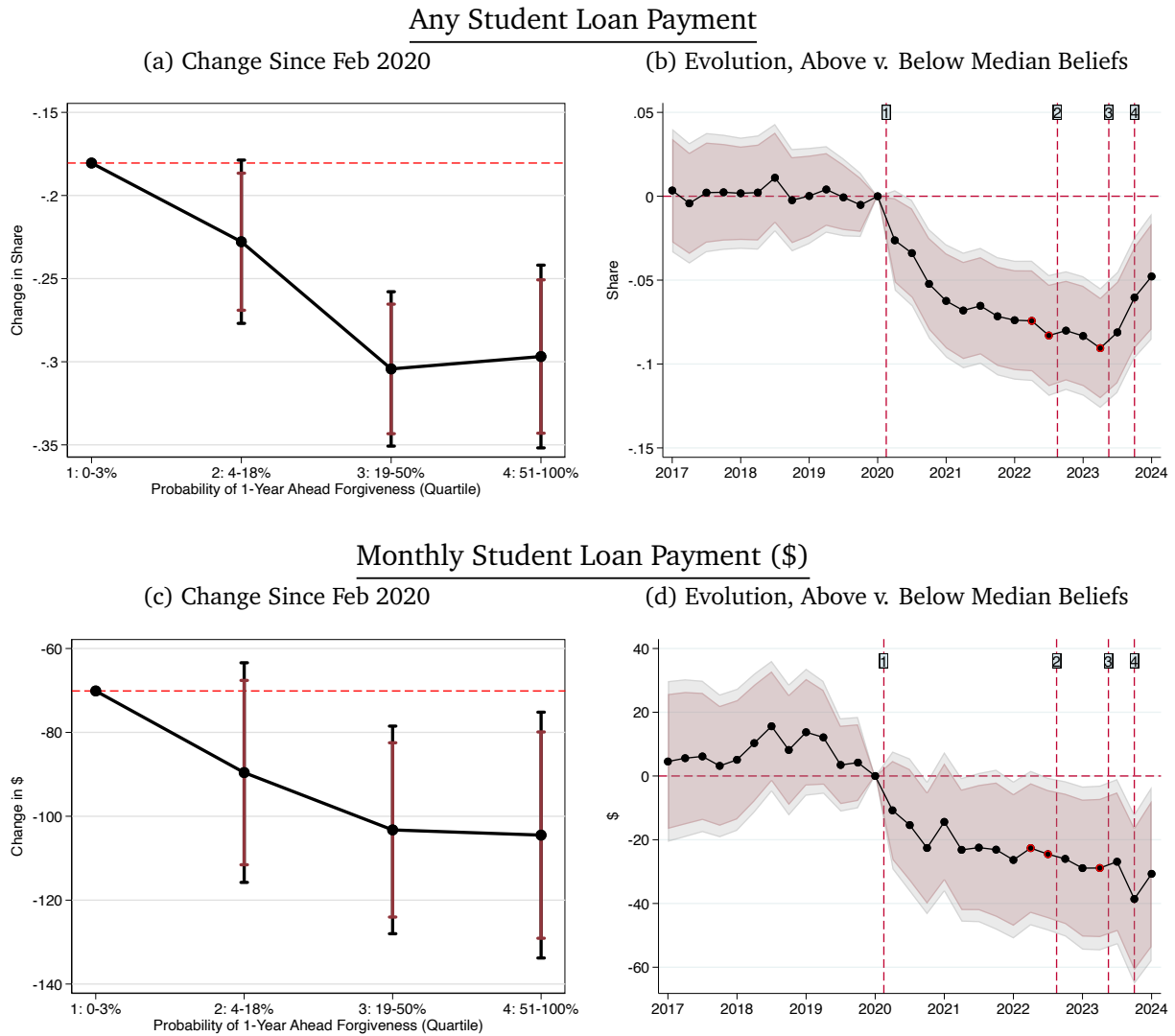


(b) Full Distribution, Selected Dates



Notes: Figure shows beliefs about the repayment pause being extended past the previously announced end date. Panel (a) reports mean beliefs over time, and panel (b) plots the distribution of beliefs prior to three separate extensions observed in our sample period. Source: Qualtrics survey.

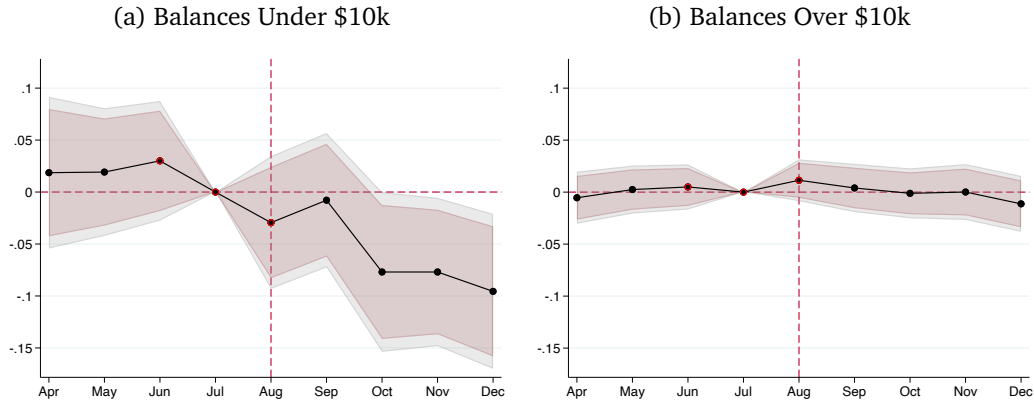
Figure 5: Evolution of Student Loan Payment, By 1-Year-Ahead Forgiveness Beliefs



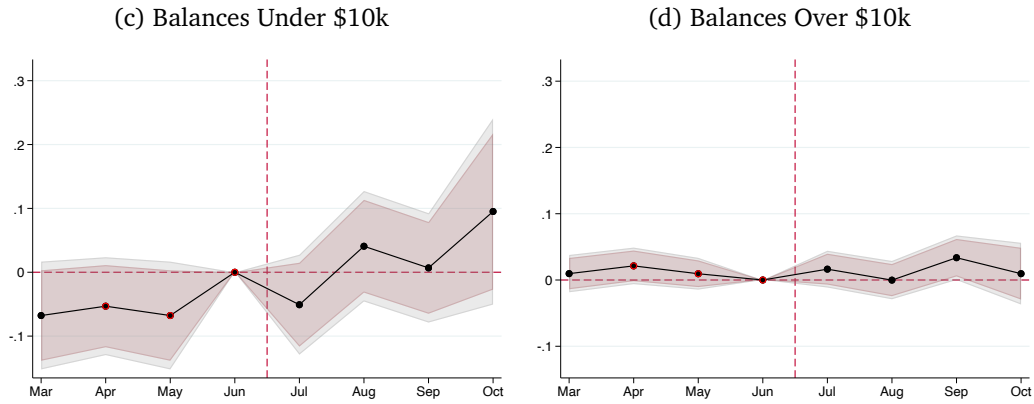
Notes: Left-hand panel reports the change in the given outcome at the time of first survey and Feb 2020, by quartiles of beliefs. Outcomes are regression adjusted to controls for month surveyed, state, whether an individual made any student loan payments in 2019, had any mortgage in 2019, zipcode-level housing price changes between February 2020 and survey date, and county-level COVID cumulative mortality rates at survey date. 95 (black) and 90 (maroon) percent confidence intervals are shown. Right-hand panel plots coefficients from a regression of quarterly time dummies interacted with an indicator of above-median one-year ahead forgiveness beliefs at time of survey. Coefficients are relative to 2020:Q1. Red circles denote main survey waves: June-Aug. 2022, and Apr.-June 2023. Regression also includes controls for state time trends, survey wave time trends, in repayment in 2019 time trends, any mortgage in 2019 time trends, zipcode-level housing prices, and county-level cumulative COVID mortality. September 2022 is recoded in 2022:Q4 as the full month occurs post the August 24, 2022 forgiveness announcement. Standard errors clustered at the individual level. Shading denotes 95 (gray) and 90 (maroon) percent confidence intervals. Key events indicated in vertical lines: (1) Payment pause begins (2) broad-based forgiveness announced (3) Supreme Court decision finds broad-based forgiveness unconstitutional (4) end of payment pause.

Figure 6: Any Student Loan Payment Around Major Forgiveness Announcements

August 2022 Forgiveness Announcement, Pessimists About Forgiveness Relative to Optimists



June 2023 Supreme Court Decision, Optimists About Forgiveness Relative to Pessimists

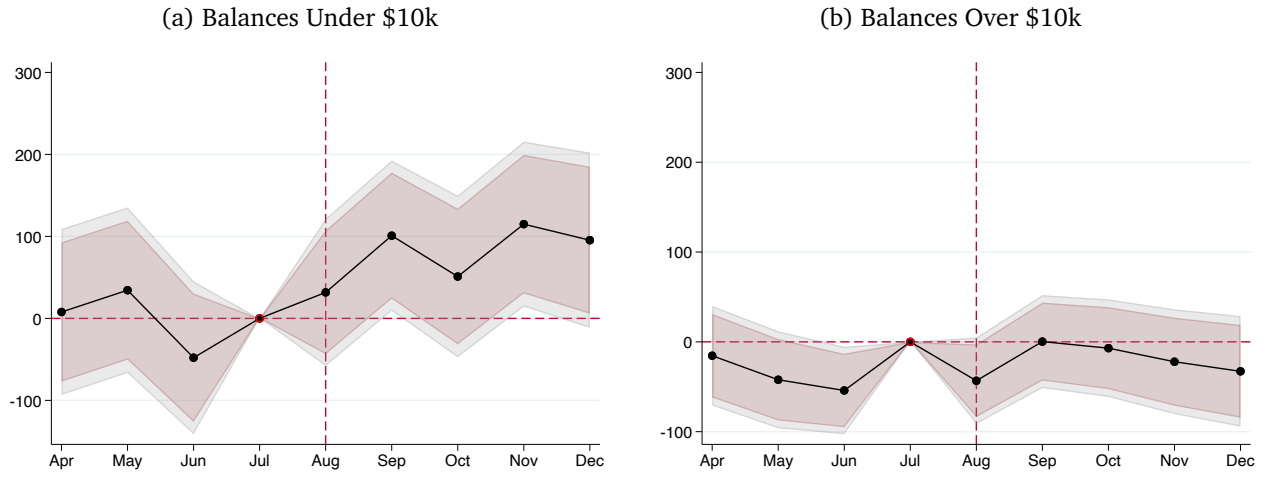


Notes: Figure plots β_k coefficients from estimating:

$$y_{i,t} = \sum \beta_k \mathbb{1}\{t \in k\} \times \mathbb{1}\{Beliefs^{pre}\} + \alpha_i + \alpha_t + \epsilon_{i,t}$$

where $Beliefs^{pre}$ is either an indicator for having below-median 1-year ahead forgiveness beliefs prior to the August 2022 forgiveness announcement (top panel), or above-median beliefs prior to the Supreme Court decision (bottom panel). Left-hand panel shows results for borrowers with pre-period balances of \$10,000 or less, and right-hand panel shows results for borrowers with pre-period balances greater than \$10,000. Red dots indicate months beliefs were surveyed. Standard errors clustered at the individual level. Shading denotes 95 (gray) and 90 (maroon) percent confidence intervals. Source: Credit data.

Figure 7: Numerator Spending Around August 2022 Forgiveness Announcement, Pessimists About Forgiveness Relative to Optimists



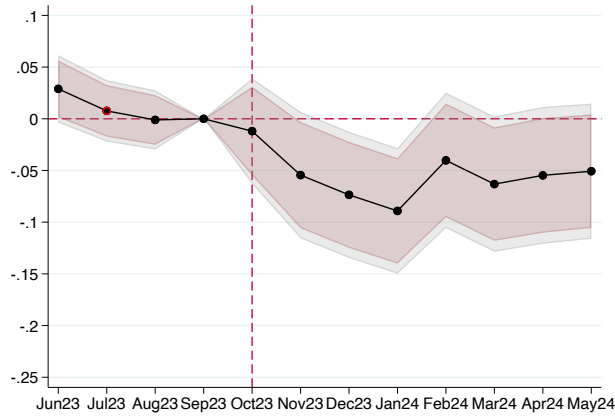
Notes: Figure plots β_k coefficients from estimating:

$$y_{i,t} = \sum \beta_k \mathbb{1}\{t \in k\} \times \mathbb{1}\{Beliefs^{pre}\} + \alpha_i + \alpha_t + \epsilon_{i,t}$$

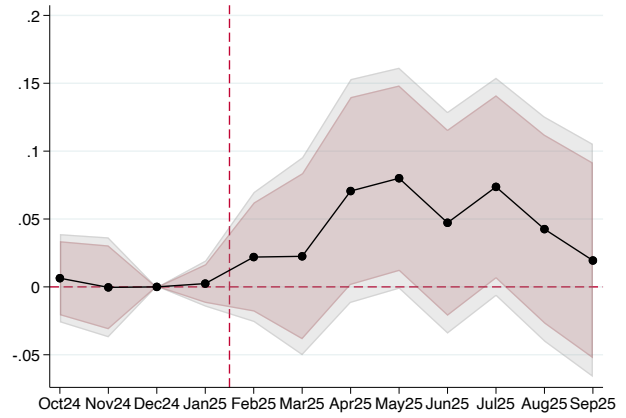
where $Beliefs^{pre}$ is an indicator for having below-median 1-year ahead forgiveness beliefs prior to the August 2022 forgiveness announcement. Left-hand panel shows results for borrowers with pre-period balances of \$10,000 or less, and right-hand panel shows results for borrowers with pre-period balances greater than \$10,000. Red dots indicate months beliefs were surveyed. Standard errors clustered at the individual level. Shading denotes 95 (gray) and 90 (maroon) percent confidence intervals. Source: Numerator data.

Figure 8: Student Loan Payments and Delinquency Around End of Payment Pause and On-Ramp, Optimists Relative to Pessimists

(a) Any Student Loan Payment, End of Payment Pause



(b) 90 Days Past Due, End of On-Ramp to Repayment

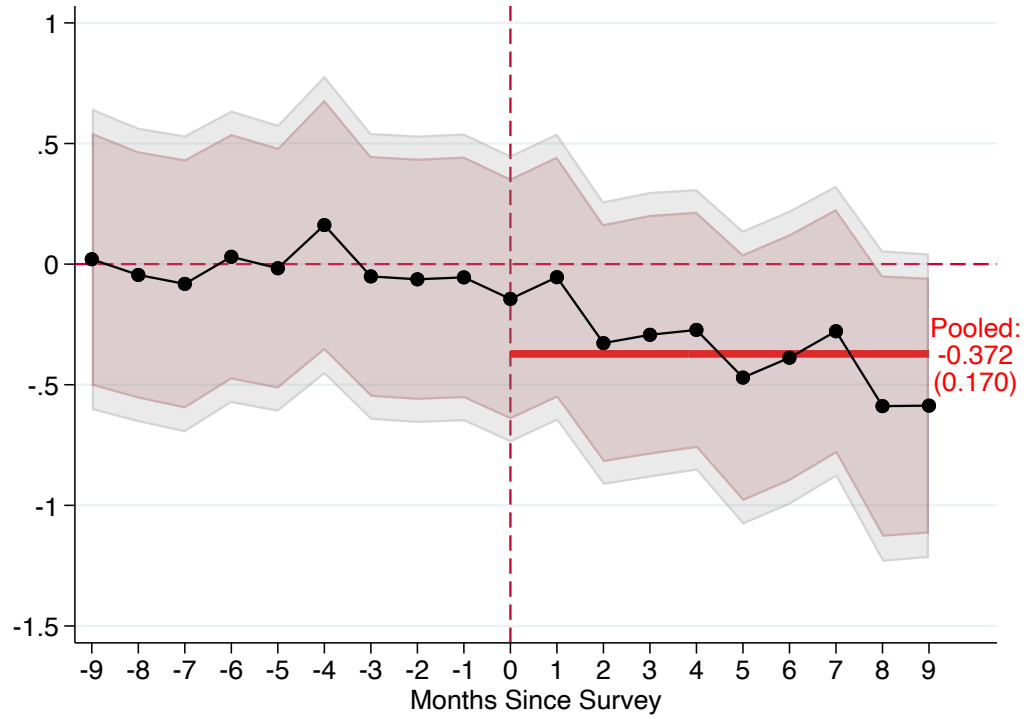


Notes: Figure plots β_k coefficients from estimating:

$$y_{i,t} = \sum \beta_k \mathbb{1}\{t \in k\} \times \mathbb{1}\{Optimist^{Pre}\} + \alpha_i + \alpha_t + \epsilon_{i,t}$$

where $Optimist^{Pre}$ is an indicator for above median beliefs about the repayment pause being extended beyond October 2023. Coefficients in Panel (a) are relative to September 2023, and coefficients in Panel (b) are relative to December 2024. Red dots indicate months beliefs were surveyed. Standard errors clustered at the individual level. Shading denotes 95 (gray) and 90 (maroon) percent confidence intervals. Source: Credit data.

Figure 9: Any Student Loan Payment and Beliefs, Instrumental Variable Estimates



Notes: Figure reports the effect of a 1 percentage point increase in beliefs on the propensity to make a student loan payment. Figure plots separate IV estimates from running the following IV regression, where the outcome is defined as a different month around the date of the information treatment:

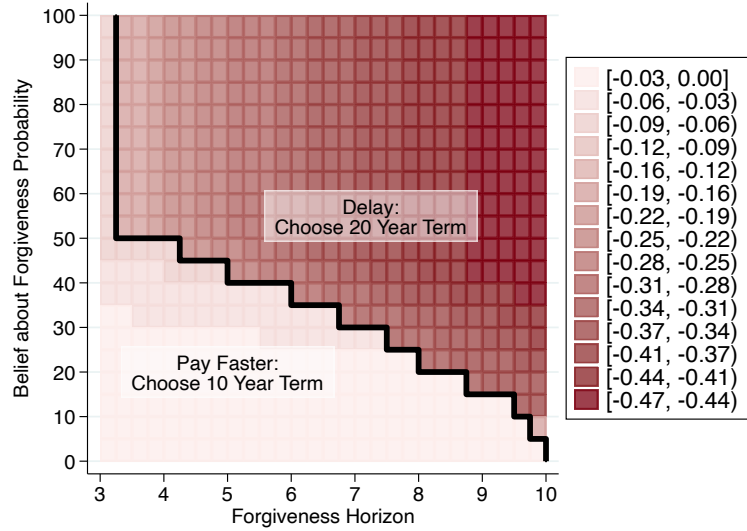
$$y_{it}^h = \beta_{IV}^h \widehat{\pi}^1 + \delta^h X_i + \epsilon_{it}^h$$

$$\pi_{it}^1 = b_1 \mathbb{1}\{OptiTreat\} + d'X_i + u_{it}$$

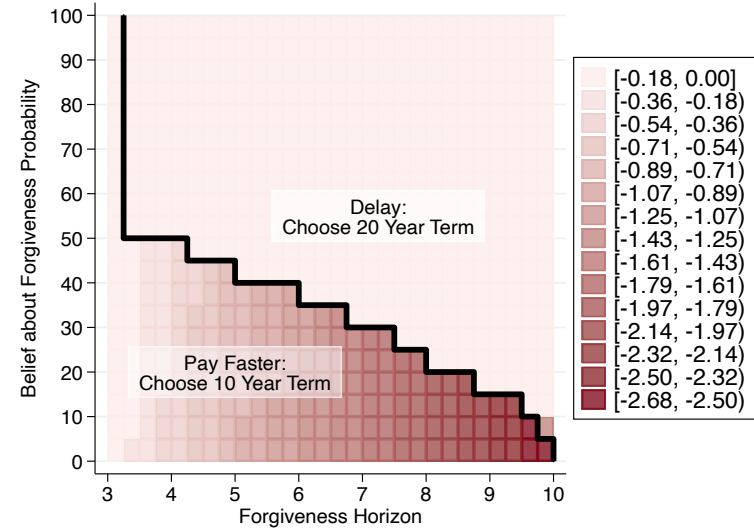
“Pooled” estimate from Table 6. In all regressions, the control group received no information treatment or the pessimistic treatment. Some individuals are in multiple independent treatments. Standard errors clustered at the individual level. Source: credit data.

Figure 10: Equivalent Variation (%) Between Ex-Post Correct Belief and Various Ex-Ante Beliefs

(a) Equivalent Variation: If No Forgiveness



(b) Equivalent Variation: If Forgiveness



Notes: Figure answers the question: “What constant percent of my consumption would I need to give up under ex-post correct beliefs [Beliefs of 0% in Panel (a) or 100% forgiveness by the year indicated on the X-axis in Panel (b)], to make utility equivalent to having the beliefs indicated on the y-axis and the forgiveness horizon given on the x-axis.” Agents choose an optimal term choice (10 or 20 years), based on their ex-ante beliefs. The thick black line denotes the optimal term choice given the ex-ante beliefs about forgiveness for the horizon indicated on the X-axis. Forgiveness refers to receiving forgiveness on remaining balance. In our baseline calibration, agents are assumed to have CRRA preferences with CRRA parameter $\rho = 2$ and discount factor $\beta = 0.96$. The rate of borrowing for student loans, r_b , is set at 6.5 percent; the rate for saving and outside borrowing, r_s , is set at 5 percent. The first three years payments are assumed paused and the loan accrues no interest. Interest begins to accrue after three years from origination. Agents are otherwise assumed to be identical across beliefs and loan characteristics. Student loan balances, income and beliefs are calibrated based on credit bureau data and survey responses.

Tables

Table 1: Summary Statistics, by Above/Below Median Beliefs

	(1) “Pessimistic” Beliefs ∈ [0,18]	(2) “Optimistic” Beliefs ∈ [19,100]	(3) Difference (2)-(1)
Student Loan Outcomes			
Mean balance	70,021.52	72,148.61	2,127.08
Median balance	46,605.00	49,768.00	3,163.00
Newest, age in years	6.43	6.10	-0.33
Any deferred	0.31	0.34	0.03
Any delinquent	0.06	0.06	0.00
Any monthly payment	0.49	0.43	-0.05***
Mean monthly payment if >0	410.50	361.26	-49.24*
Other Credit and Employment Outcomes			
Any revolving balance	0.84	0.82	-0.01
Mean revolving balance if >0	10,909.25	10,824.99	-84.26
Any auto balance	0.60	0.61	0.01
Mean auto balance if >0	19,284.78	19,340.18	55.40
Any mortgage balance	0.41	0.44	0.03*
Mean mortgage balance if >0	194,959.21	176,556.62	-18,402.59***
Vantage 4.0 Credit Score	665.79	657.11	-8.68**
Any Employment	0.315	0.316	0.001
Gross monthly income if >0	7,112.48	6,571.41	-541.07
Borrower Characteristics			
Age	45.56	46.78	1.22***
Female	0.59	0.57	-0.02
Married	0.48	0.48	-0.01
Has children	0.50	0.46	-0.04*
Zipcode and County Characteristics			
Zipcode AGI (2019)	67,443.59	64,980.32	-2,463.27**
Zipcode ZHVI Home prices (Feb 2020)	252,294.54	245,803.48	-6,491.06
Zipcode Δln(ZHVI), Dec 2023-Feb 2020	0.34	0.33	-0.00
County COVID mortality cumul. per 100k	312.58	319.62	7.04*
N	1467	1427	

Notes: Table shows summary statistics comparing those with below median to above median beliefs about 1-year ahead forgiveness as measured at the time of first survey. Restricted to those with balance reported in credit bureau data as of survey month. Outcomes shown for February 2020 (pre-COVID payment pause) unless otherwise specified. Stars denote statistical significance of differences in means using robust standard errors, * p<0.1, ** p<0.05, *** p<0.01. Source: Zipcode AGI from IRS Tax Stats; ZHVI from Zillow Research; COVID mortality from Opportunity Insights. Other outcomes from credit and employment data.

Table 2: Student Loan Payment Around Major Forgiveness Announcements

	(a) 8/2022 Executive Forgiveness Order			(b) 6/23 Supreme Court Decision		
	(1)	(2)	(3)	(4)	(5)	(6)
	Prob Forgive	Any Pay	Pay (\$)	Prob Forgive	Any Pay	Pay (\$)
Beliefs (Below/Above Median) × Post	0.202*** (0.0690)	-0.00137 (0.00925)	1.489 (4.935)	-0.267*** (0.0277)	0.00474 (0.0130)	-5.388 (8.629)
Post × Bal<\$10k	0.0869 (0.128)	0.00897 (0.0163)	0.00897 (8.633)	-0.0584** (0.0257)	0.00877 (0.0254)	0.00877 (9.954)
Beliefs × Post × Bal<\$10k	-0.0694 (0.176)	-0.0775*** (0.0274)	-38.47** (16.15)	0.0907 (0.152)	0.0654* (0.0376)	13.89 (11.46)
Pre-Period Mean	0.21	0.22	80.25	0.26	0.24	89.44
Individuals	143	1,776	1,776	332	944	944
Total Observations	286	14,208	14,208	664	7,545	7,545
Individual FE	X	X	X	X	X	X
Time FE	X	X	X	X	X	X

Notes: Table reports coefficients from estimating:

$$y_{i,t} = \beta_1 \text{Beliefs}^{pre} \times \text{Post}_t + \beta_2 \mathbb{1}\{\text{Balance}^{pre} < \$10k\} \times \text{Post}_t + \beta_3 \text{Beliefs}^{pre} \times \mathbb{1}\{\text{Balance}^{pre} < \$10k\} \times \text{Post}_t + \alpha_i + \alpha_t + \epsilon_{i,t}$$

In column (1) and (4), regression is restricted to individuals in a post-event resurvey of beliefs. In Panel (a), the estimation window is from April 2022-December 2022, excluding August 2022 (partial treatment). Post_t refers to period from September 2022, the first full month following the 8/24/2022 forgiveness announcement. Beliefs^{pre} is an indicator for having below-median beliefs prior to the announcement (i.e. having “Pessimistic” beliefs). In panel (b), the estimation window is from March 2023-October 2023. Post_t refers to period from July 2023, following the June 30 Supreme Court decision, and Beliefs^{pre} is an indicator for having above-median beliefs prior to the Supreme Court decision (i.e. having “Optimistic” beliefs). Standard errors clustered at the individual level in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Source: Credit data.

Table 3: Spending Responses Around August 2022 Forgiveness Announcement

	(1)	(2)	(3)	(4)	(5)
	Total Spend	Food	All Other	Ln Total	Ln Food
Beliefs (Below Median) $\times$ Post	12.49 (18.30)	4.689 (7.281)	10.23 (15.59)	0.0127 (0.0205)	0.0271 (0.0284)
Post $\times$ Bal < 10k	-34.07 (28.52)	10.99 (10.61)	-38.21 (24.73)	-0.0371 (0.0337)	0.0686 (0.0468)
Beliefs $\times$ Post $\times$ Bal < 10k	80.48** (38.64)	-4.826 (14.56)	74.47** (33.64)	0.0894** (0.0442)	-0.0362 (0.0615)
Pre-Period Mean	1127.97	324.90	796.13	6.77	5.38
Individuals	3028	3028	3028	3028	2981
Observations	23975	23975	23975	23975	21761
Individual FE	X	X	X	X	X
Time FE	X	X	X	X	X

Notes: Table reports coefficients from estimating:

$$y_{i,t} = \beta_1 \text{Beliefs}^{pre} \times \text{Post}_t + \beta_2 \mathbb{1}\{\text{Balance}^{pre} < \$10k\} \times \text{Post}_t + \beta_3 \text{Beliefs}^{pre} \times \mathbb{1}\{\text{Balance}^{pre} < \$10k\} \times \text{Post}_t + \alpha_i + \alpha_t + \epsilon_{i,t}$$

Estimation window from April 2022-December 2022, excluding August 2022 (partial treatment). Post_t refers to period from September 2022, the first full month following the 8/24/2022 forgiveness announcement. Beliefs^{pre} is an indicator for having below-median beliefs prior to the announcement (i.e. having “Pessimistic” beliefs). Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Source: Numerator data.

Table 4: Student Loan Payments and Delinquency Around End of Payment Pause and End of On-Ramp to Repayment

	(a) 10/2023 Resumption of Payments Due				(b) 1/2025 Delinquency Reporting	
	(1)	(2)	(3)	(4)	(5)	(6)
	Any Pay	Pay (\$)	Any Pay	Pay (\$)	90 Days Past Due	90 Days Past Due
Optimist (Pause) × Post	-0.0628** (0.0255)	-39.77*** (12.27)	-0.0714*** (0.0274)	-40.32*** (12.36)	0.063* (0.037)	0.062 (0.039)
Optimist (Forgiveness) × Post			0.0256 (0.0274)	1.630 (12.47)		0.005 (0.040)
Pre-Period Mean	0.22	89.47	0.22	89.47	0.026	0.026
Individuals	773	773	773	773	452	452
Total Observations	6,956	6,956	6,956	6,956	2,129	2,129
Individual FE	X	X	X	X	X	X
Time FE	X	X	X	X	X	X

Notes: Columns (1), (2) and (5) report coefficients from estimating:

$$y_{i,t} = \beta_1 \mathbb{1}\{\text{Optimist}^{pre}\} \times \text{Post}_t + \alpha_i + \alpha_t + \epsilon_{i,t}$$

where Optimist^{pre} is an indicator for above-median beliefs about the payment pause extension, as collected post the June 30 Supreme Court decision. Columns (3), (4) and (6) also include above-median forgiveness beliefs collected at this time. Sample is restricted to student loan borrowers with beliefs surveyed post the June 30 Supreme Court Decision. Columns (5)-(6) additionally restrict to borrowers with non-zero balance due. Estimation window for Columns (1)-(4) is from June 2023-February 2024, and Post_t refers to the period from October 2023 when payments were again due. Estimation window for Columns (5)-(6) is October 2024-December 2024 (2024Q4) and April 2025-June 2025 (2025Q2). Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Source: Credit data.

Table 5: Effect of Information Treatment on Beliefs

	(1)	(2)	(3)	(4)	(5)
	Prob Forgive	Prob Forgive	Prob Forgive	Prob Forgive	Prob Forgive
Opti Treat	0.0365*** (0.00877)	0.0375*** (0.0100)		0.0434*** (0.0111)	0.0411*** (0.0118)
Pessi Treat		0.00195 (0.00978)			
Opti Treat-1			0.0337*** (0.0106)		
Opti Treat-2			0.101 (0.0795)		
Opti Treat-3			0.0288* (0.0163)		
Opti Treat $\times$ Bal <10k					0.0187 (0.0340)
Bal <10k					-0.0107 (0.0205)
Observations	5128	5128	5128	3226	3226
R^2	0.0298	0.0298	0.0293	0.0360	0.0361
F-stat	17.32	8.683	4.881	15.45	5.245
Survey Year-Month FE	X	X	X	X	X
Pre Balance>0				X	X
No Payment Required				X	X

Notes: Regression of 1-year ahead forgiveness beliefs, on the experimental information treatment indicated in the rows. The text of the information treatment is provided in Appendix B. In all regressions, there is a control group which received no information treatment. Some individuals are in multiple independent treatments. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Source: Credit data.

Table 6: Effect of Information Treatment on Student Loan Payments and Voting Intentions

	(1)	(2)	(3)
	Student Loans		Prob Vote
	Any Pay	Pay (\$)	≥ 0.5
Prob Forgive $\times$ Post Treat	-0.372** (0.170)	-220.1 (179.2)	
Post Treat	0.122** (0.0545)	67.63 (56.92)	
Prob Forgive			1.143** (0.558)
Observations	25791	25791	2359
Individual FE	X	X	
Year-Month FE	X	X	X
Pre Balance > 0	X	X	X
No Payment Required	X	X	X

Notes: Columns (1)-(2) report results from an IV event-study framework:

$$y_{it} = \beta_{IV} \pi^1 \times \widehat{\mathbb{1}\{Post\}} + \beta_2 \mathbb{1}\{Post\} + \alpha_i + \alpha_t + \epsilon_{it}$$

$$\pi_{it}^1 \times \mathbb{1}\{Post\} = b_1 \mathbb{1}\{OptiTreat\} \times \mathbb{1}\{Post\} + b_2 \mathbb{1}\{Post\} + a_i + a_t + u_{it}$$

where $\mathbb{1}\{Post\}$ is a post survey indicator, π^1 are one-year ahead beliefs about forgiveness, $\mathbb{1}\{OptiTreat\}$ is an indicator for receiving the optimist information treatment, and α_i and α_t are individual and time fixed effects. The event window is 4 months around the survey month, excluding the survey month due to partial treatment. Column (3) examines the simple IV regression, restricting only to the survey month:

$$y_{it} = \beta_{IV} \widehat{\pi^1} + \alpha_t + \epsilon_{it}$$

$$\pi_{it}^1 = b_1 \mathbb{1}\{OptiTreat\} + a_t + u_{it}$$

In all regressions, the control group received no information treatment or the pessimistic treatment. Some individuals are in multiple independent treatments in the event studies. Standard errors clustered at the individual level in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01. Source: Credit data.

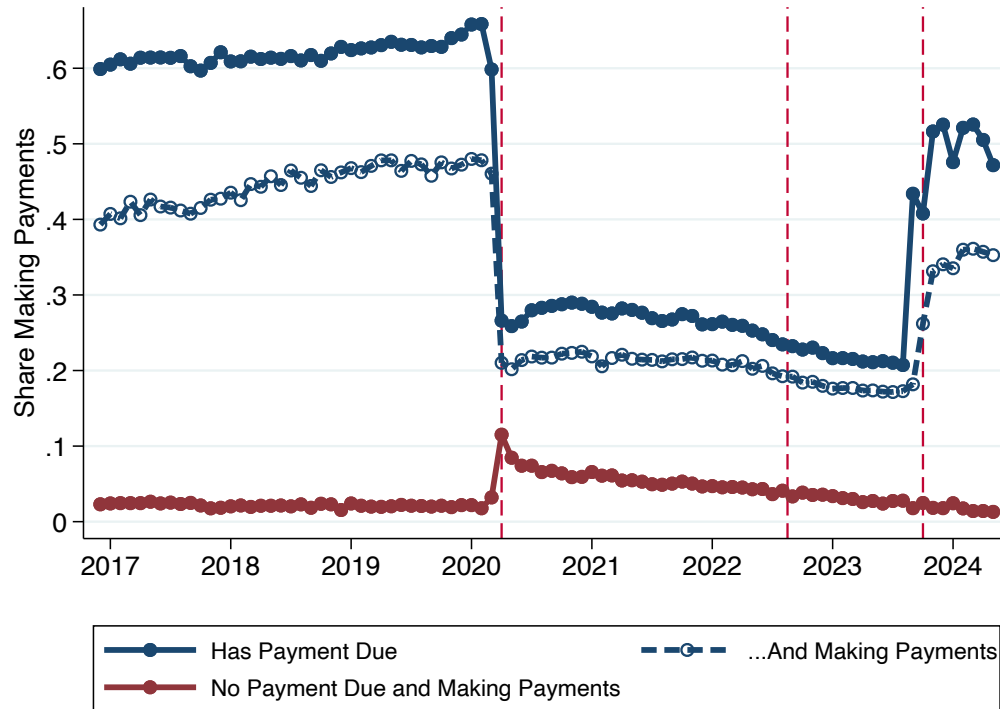
Table 7: Present Value of Lifetime Willingness to Pay as Percent of Initial Loan Balance

	(1)	(2)	(3)	(4)	(5)
	Fixed Within Repayment Plan			Optimal Repayment Plan	
	Forgiveness Realizations:			Forgiveness Realizations:	
	No Forgiveness	Forgiveness 3 Years from Origination	Forgiveness 10 Years from Origination	No Forgiveness	Forgiveness 10 Years from Origination
Choose 10-year repayment plan:					
0% Forgiveness	0.00	-1.30	0.00	0.00	-43.85
50th Percentile	0.00	-0.81	0.00	0.00	-43.85
Mean	-0.15	-0.65	0.00	-0.15	-43.85
100%	-1.23	0.00	0.00	-1.23	-43.85
Choose 20-year repayment plan:					
0% Forgiveness	0.00	-1.47	-1.74	-5.35	-1.74
50th Percentile	0.00	-0.98	-1.10	-5.51	-1.10
Mean	-0.15	-0.81	-0.95	-5.51	-0.95
100%	-1.38	0.00	0.00	-6.88	0.00

Notes: This table shows the present value of the lifetime willingness to pay, expressed as a percent of the initial loan balance. For each scenario, we first calculate the equivalent variation as the constant share of consumption the borrower would need to give up under ex-post correct beliefs to make utility equivalent to the indicated beliefs (this equivalent variation is reported in Appendix Table A.6). We then calculate the present value discounted at the savings rate (r_s), and report as a percent of the initial loan balance (B_0). Columns (1)-(3) compare differences to correct beliefs within a payment plan choice (10-year or 20-year). In columns (4)-(5), borrowers choose the repayment plan that maximizes their utility under their ex-ante beliefs about forgiveness. “No forgiveness” is relative to a counterfactual where forgiveness is expected after 3 years at the indicated beliefs. “Forgiveness” refers to receiving forgiveness on remaining balance. In our baseline calibration, borrowers are assumed to have CRRA preferences with CRRA parameter $\rho = 2$ and discount factor $\beta = 0.96$. The rate of borrowing for student loans, r_b , is set at 6.5 percent; the rate for saving and outside borrowing, r_s , is set at 5 percent. The first three years payments are assumed paused and the loan accrues no interest. Agents are otherwise assumed to be identical across beliefs and loan characteristics. Student loan balances, income and beliefs are calibrated based on credit bureau data and survey responses. Under this calibration, the 10-year repayment plan is optimal if no forgiveness is expected, and 20-year repayment plan is optimal if forgiveness is expected. Ex-post correct beliefs are in bold.

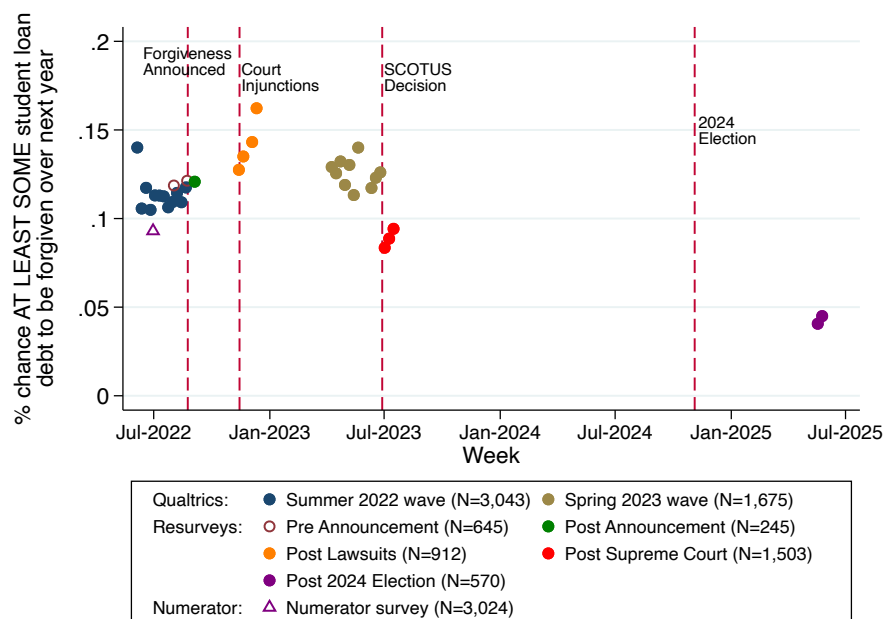
A Appendix Figures and Tables

Figure A.1: Share Making Any Student Loan Payment-Respondents



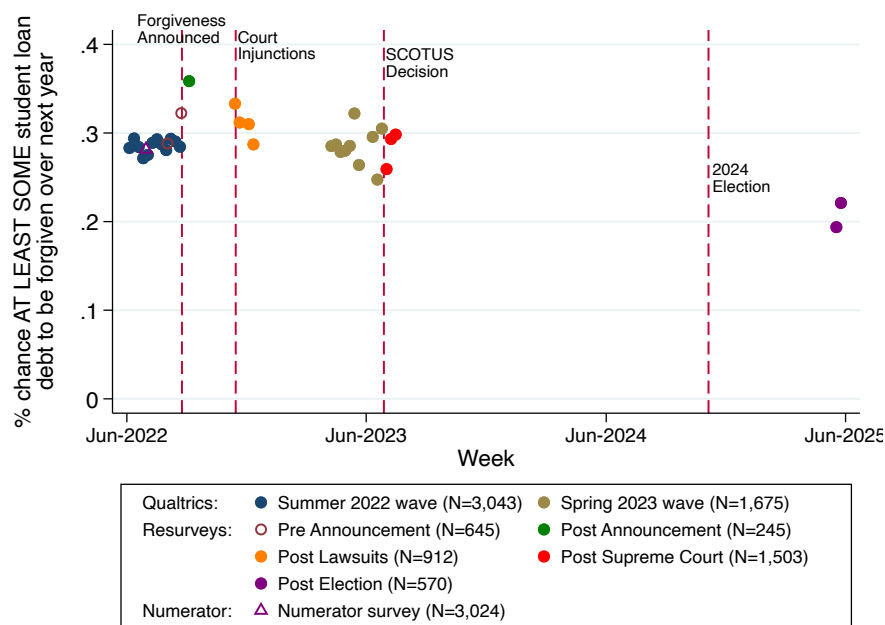
Notes: Figure shows the share of borrowers making a payment on student loans, conditional on having an open balance in February 2020. Vertical lines indicate April 2020, the start of the payment pause, August 2022 Biden's Forgiveness Announcement, and October 2023, the month the payment pause ended. Source: Credit data.

Figure A.2: “Subjective Uncertainty”: Second Moment of Survey Expectations Over Time



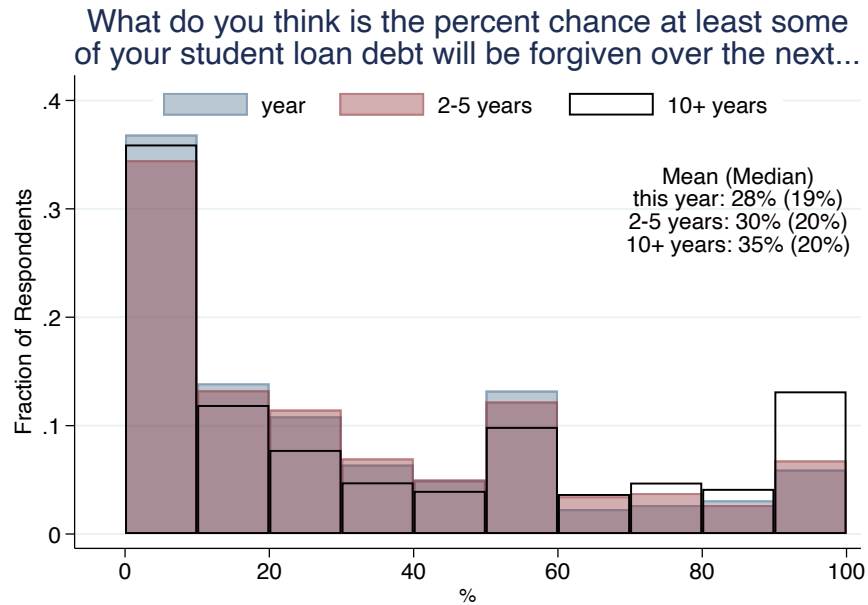
Notes: Figure shows the second moment of beliefs about 1-year ahead forgiveness from our survey. Each dot is the mean of second moment beliefs over individuals surveyed in the week. Source: Qualtrics survey and Numerator survey.

Figure A.3: “Policy Disagreement”: Standard Deviation of Survey Expectations Over Time



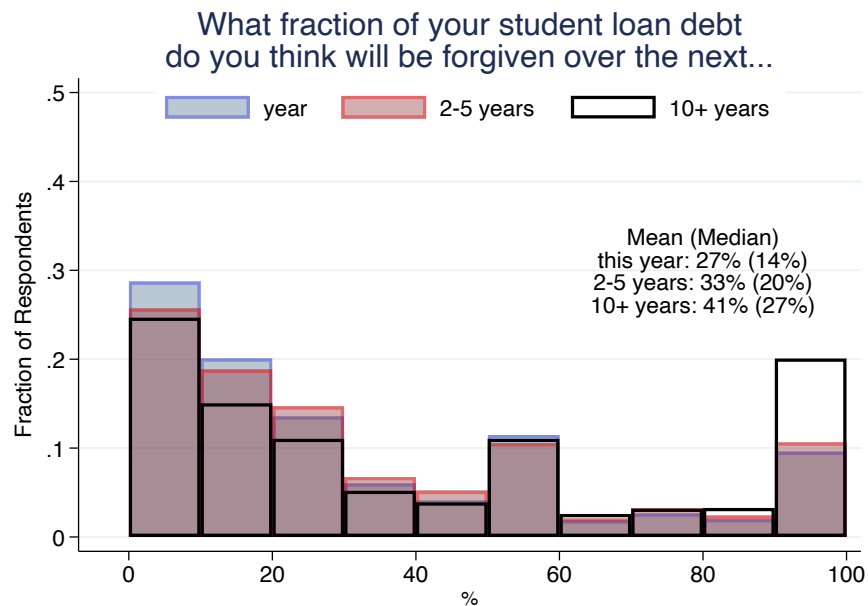
Notes: Figure shows cross-sectional standard deviation of beliefs about 1-year ahead forgiveness from our survey. Each dot is the standard deviation of beliefs over individuals surveyed in the week. Source: Qualtrics survey and Numerator survey.

Figure A.4: Full Distribution of Forgiveness Beliefs at Various Horizons



Notes: Figure shows expectations about receiving any forgiveness over various forgiveness horizons. Beliefs surveyed June 2022-July 2023.

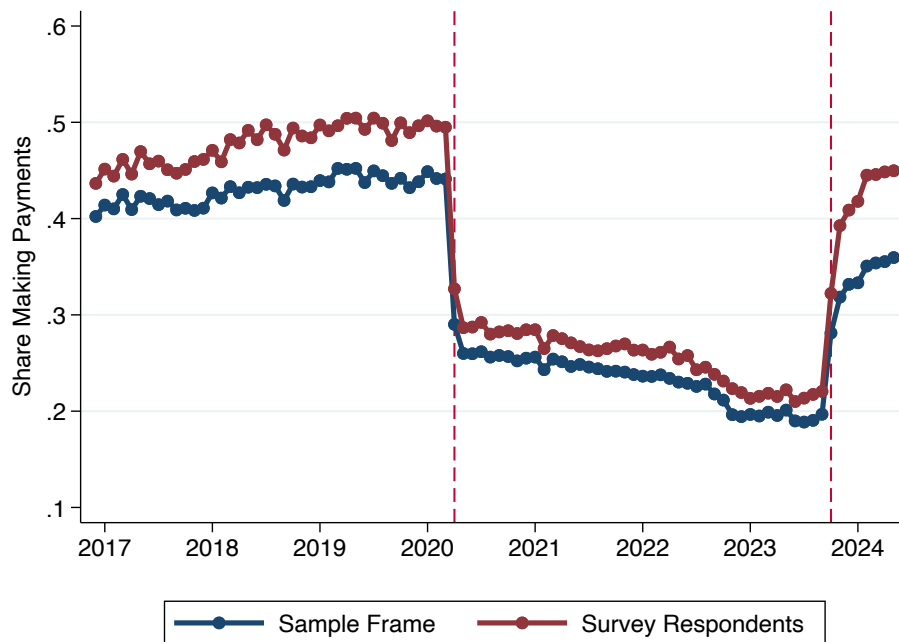
Figure A.5: Fraction of Student Loan Debt Expected to Be Forgiven (Positive Beliefs)



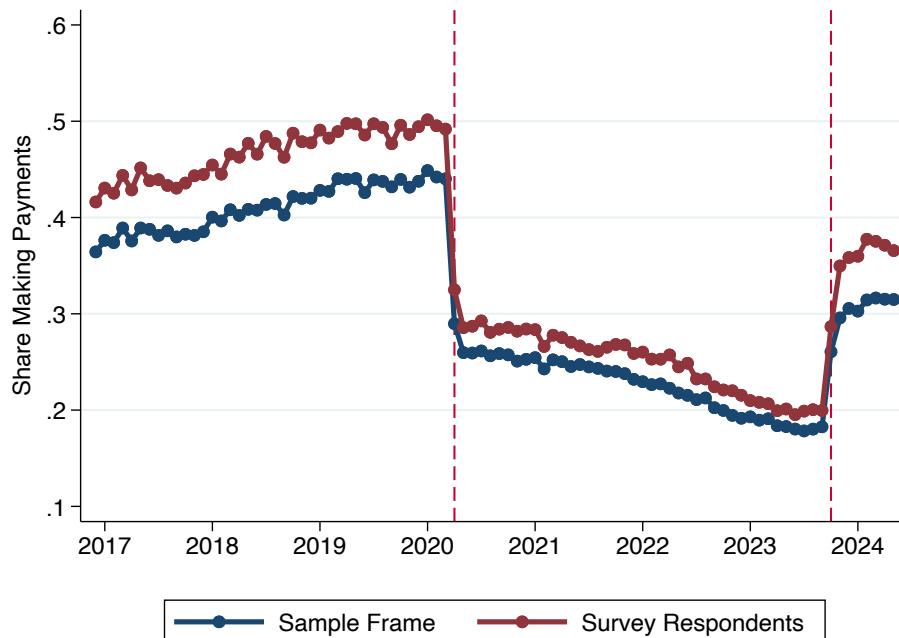
Notes: Figure shows the fraction of student loan debt respondents expect to be forgiven over various forgiveness horizons, among respondents with positive beliefs about forgiveness. Beliefs surveyed June 2022-July 2023.

Figure A.6: Share Making Any Student Loan Payment-Respondents v Survey Frame

(a) Conditional on Open balance in Month

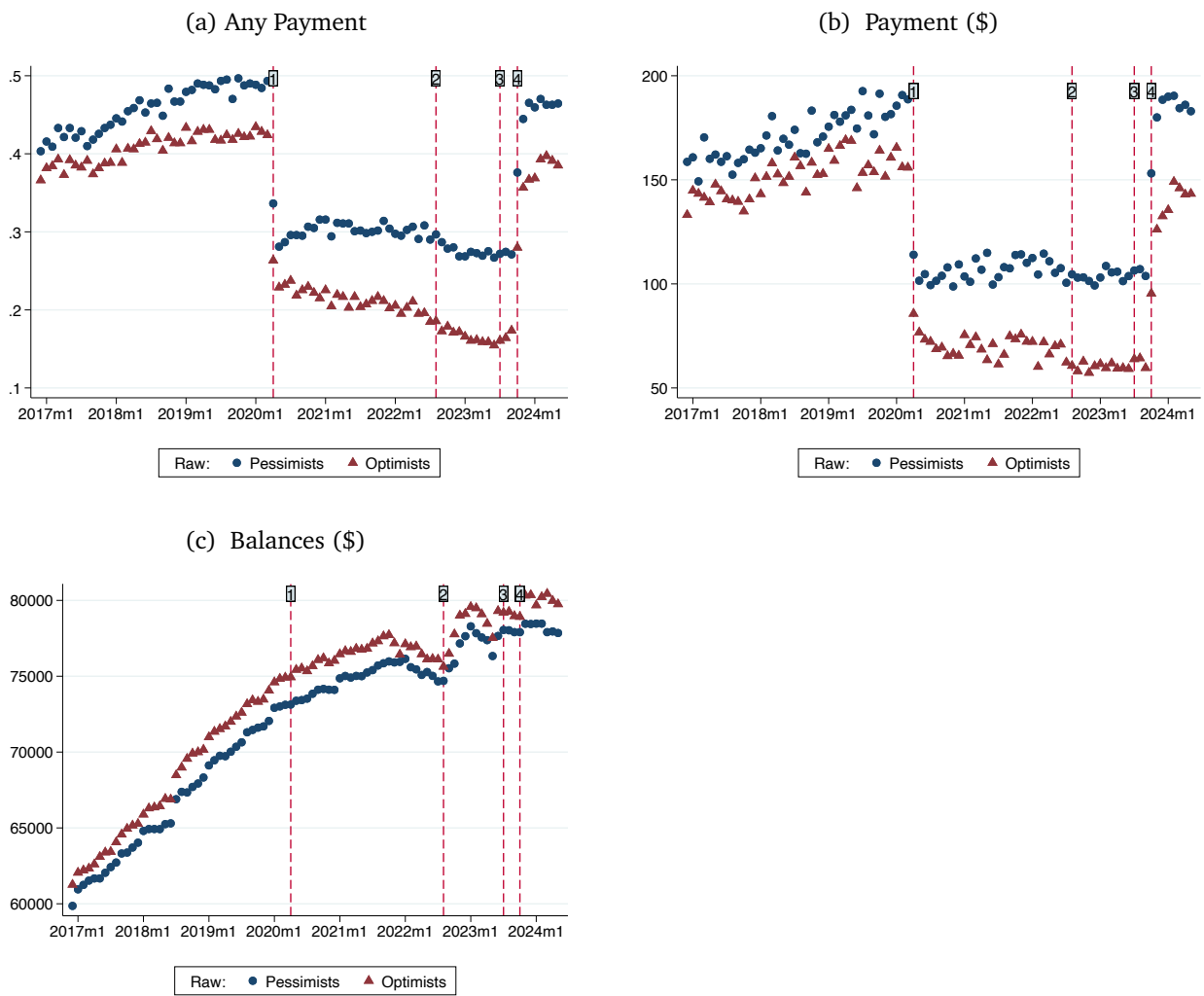


(b) Conditional on Balance in Feb 2020



Notes: Figure shows the share of borrowers making a payment on student loans. Vertical lines indicate April 2020, the start of the payment pause, and October 2023, the month the payment pause ended. Source: Credit data.

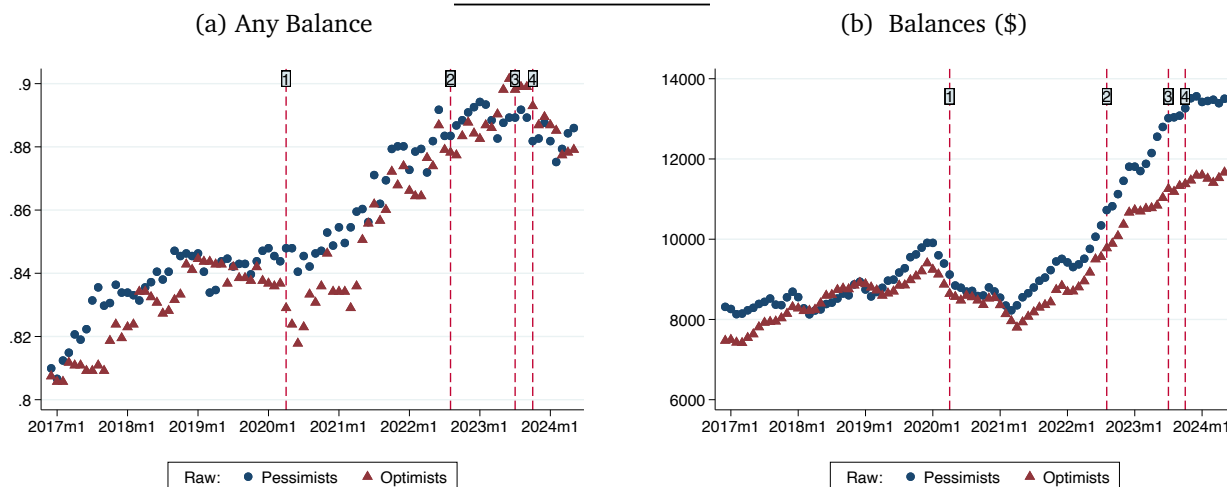
Figure A.7: Raw Means of Student Loan Outcomes, Above v. Below Median Beliefs



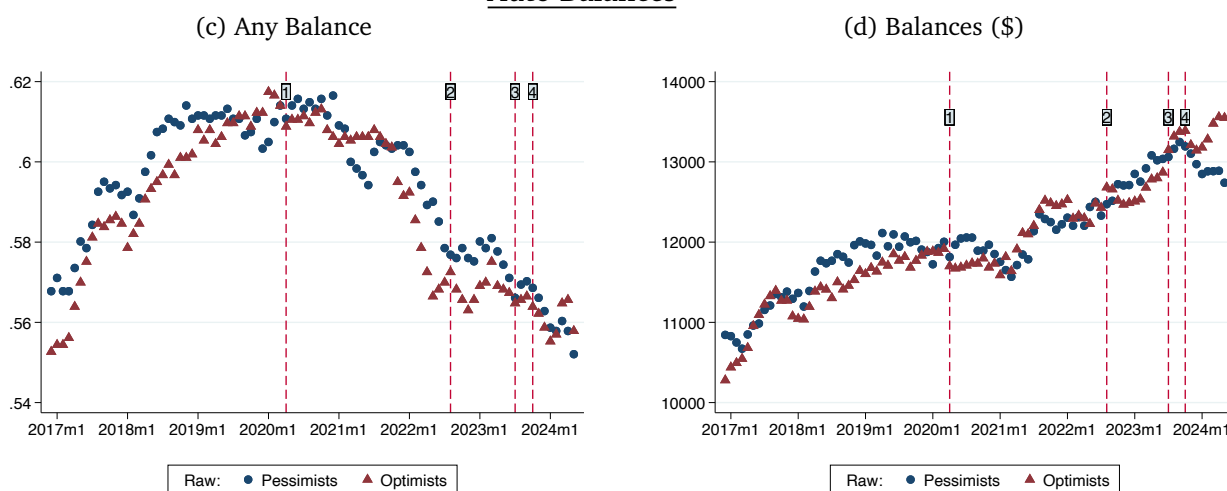
Notes: Raw means by above/below median beliefs at time of survey. Source: credit data.

Figure A.8: Raw Means of Credit Outcomes, Above v. Below Median Beliefs

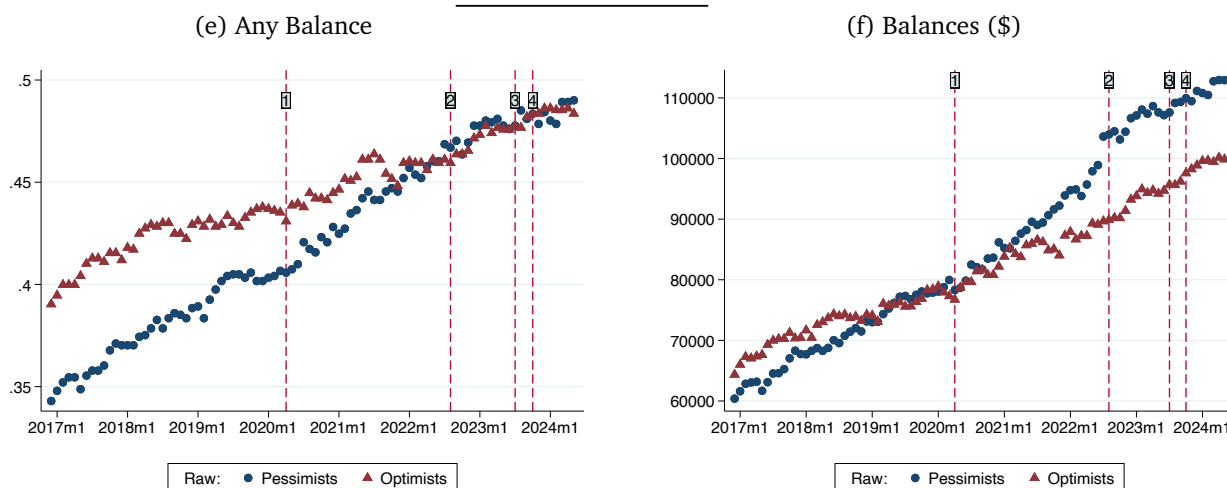
Revolving Balances



Auto Balances



Mortgage Balances



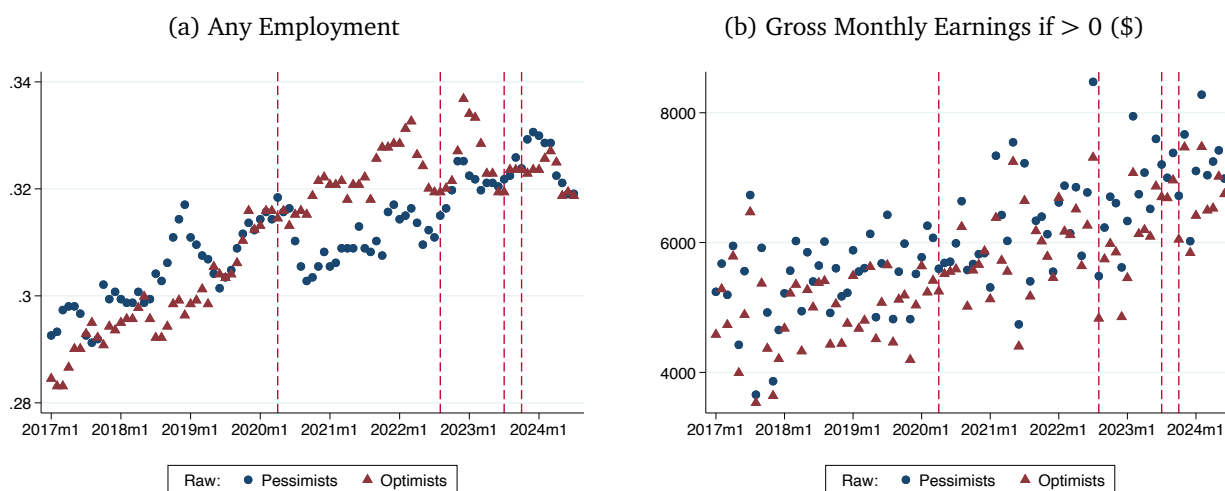
Notes: Raw means by above/below median beliefs at time of survey. Source: credit data.

Figure A.9: Raw Means of Numerator Spending Outcomes, Above v. Below Median Beliefs



Notes: Raw means by above/below median beliefs at time of survey. Source: Numerator.

Figure A.10: Raw Means of Employment Outcomes, Above v. Below Median Beliefs



Notes: Raw means by above/below median beliefs at time of survey. Source: Employment data.

Figure A.11: Evolution of Credit Outcomes, By 1-Year Ahead Forgiveness Beliefs



Notes: See notes for Figure 5. Source: Credit data.

Figure A.12: Evolution of Numerator Spending Outcomes, By 1-Year Ahead Forgiveness Beliefs

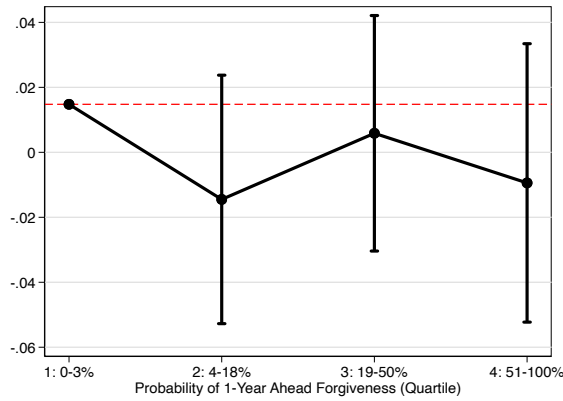


Notes: See notes for Figure 5. Source: Numerator.

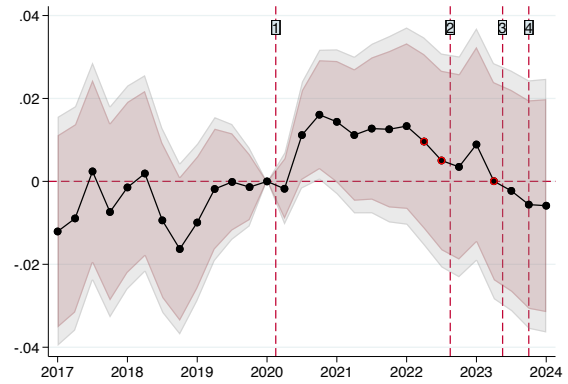
Figure A.13: Evolution of Employment Outcomes

Any Employment

(a) At Time of Survey, Change Since Feb 2020

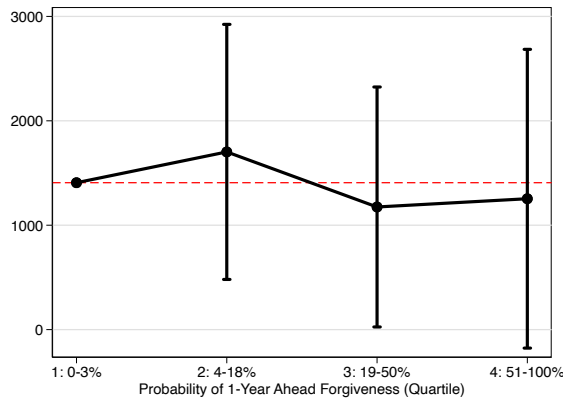


(b) Evolution, Above v. Below Median Beliefs

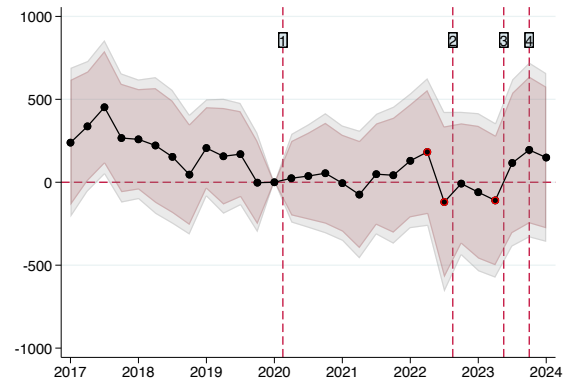


Gross Monthly Earnings if > 0 (\$)

(c) At Time of Survey, Change Since Feb 2020



(d) Evolution, Above v. Below Median Beliefs



Notes: See notes for Figure 5. Source: Employment data.

Table A.1: Consumption Survey Intentions

(a) Correlation with Beliefs				
	(1) Spend House	(2) Spend Car	(3) Other Durables	(4) Other Spend
Prob 1-Yr Forgive	0.0723** (0.0350)	0.0304 (0.0326)	0.0107 (0.0267)	0.00425 (0.0353)
Constant	0.126*** (0.0131)	0.144*** (0.0131)	0.0839*** (0.0106)	0.194*** (0.0146)
<i>N</i>	1448	1448	1448	1448
(b) Including Liquidity Controls				
	(1) Spend House	(2) Spend Car	(3) Other Durables	(4) Other Spend
Prob 1-Yr Forgive	0.0756** (0.0372)	0.00620 (0.0345)	-0.00111 (0.0291)	-0.0139 (0.0375)
Constant	0.143*** (0.0208)	0.185*** (0.0225)	0.0904*** (0.0174)	0.189*** (0.0228)
<i>N</i>	1313	1313	1313	1313
Savings Controls	X	X	X	X

Notes: Table shows a regression of intentions to increase spending in response to receiving student loan forgiveness of up to \$10,000, on one-year ahead forgiveness beliefs. Different spending categories are indicated by the column heading. Panel (b) includes controls for indicators of self-reported savings: None (\$0); \$1-4,999; \$5,000-9,999; \$10,000-14,999; \$15,000-19,999; \$20,000-29,999; \$30,000-39,999; \$40,000-49,999; \$50,000-99,999; \$100,000 or more. Respondents reporting “Don’t know” are dropped in panel b. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Summary Statistics, Survey Respondents v. Sample Frame

	(1) Respondents	(2) Sample Frame	(3) Difference (2)-(1)
Student Loan Outcomes			
Mean balance	73,104.25	49,311.30	-23,792.95***
Median balance	50,776.00	29,435.00	-21,341.00
Newest, age in years	6.30	6.28	-0.02
Any deferred	0.33	0.33	-0.00
Any delinquent	0.05	0.08	0.03***
Any monthly payment	0.50	0.45	-0.05***
Mean monthly payment if >0	390.52	351.80	-38.73**
Other Credit and Employment Outcomes			
Any revolving balance	0.84	0.78	-0.07***
Mean revolving balance if >0	11,202.09	10,356.58	-845.51**
Any auto balance	0.61	0.59	-0.02**
Mean auto balance if >0	19,333.12	19,892.85	559.73
Any mortgage balance	0.45	0.41	-0.05***
Mean mortgage balance if >0	184,149.61	186,894.47	2,744.86
Vantage 4.0 Credit Score	666.99	650.85	-16.14***
Any Employment	0.315	0.297	0.018**
Gross monthly income if >0	5,731.45	5,658.16	73.30
Borrower Characteristics			
Age	46.44	43.67	-2.77***
Female	0.58	0.51	-0.06***
Married	0.48	0.49	0.01
Has children	0.49	0.49	0.01
Zipcode and County Characteristics			
Zipcode AGI (2019)	65,864.02	66,196.98	332.96
Zipcode ZHVI Home prices (Feb 2020)	247,341.50	242,818.82	-4,522.68
Zipcode $\Delta \ln(\text{ZHVI})$, Dec 2023-Feb 2020	0.33	0.34	0.00
County COVID mortality cumul. per 100k	316.38	326.19	9.81***
Observations	2,603	26,604	

Notes: Table shows summary statistics comparing survey respondents to non-respondents. Restricted to those with student loan balance reported in credit bureau data as of February 2020. Outcomes shown for February 2020 (pre-COVID payment pause) unless otherwise specified. Stars denote statistical significance of differences in means using robust standard errors, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Source: Zipcode AGI from IRS Tax Stats; ZHVI from Zillow Research; COVID mortality from Opportunity Insights. Other outcomes from credit and employment data.

Table A.3: Credit and Employment Outcomes Around Forgiveness Announcements

(a) August 2022 Forgiveness Announcement				
	(1) Revolving Bal (\$)	(2) Auto Bal (\$)	(3) Mortgage Bal (\$)	(4) Employment Any
Pess× Post	83.91 (302.0)	700.8* (392.0)	-1363.1 (2386.3)	0.0035 (0.0098)
Post× Bal< 10k	-411.1 (537.7)	1352.1 (1083.3)	1491.2 (4344.6)	0.0048 (0.01245)
Pess× Post× Bal< 10k	-422.5 (720.0)	-2601.7** (1171.5)	606.7 (6782.8)	-0.0078 (0.0248)
Pre mean	9,620.04	12,245.57	100444.95	0.3185
N	1776	1776	1776	1776
N×T	14208	14208	14208	14208
Ind FE	X	X	X	X
Time FE	X	X	X	X

(b) June 2023 Supreme Court Decision				
	(1) Revolving Bal (\$)	(2) Auto Bal (\$)	(3) Mortgage Bal (\$)	(4) Employment Any
Opti× Post	-179.4 (370.9)	138.3 (475.3)	-2280.6 (2211.1)	0.0045 (0.0114)
Post× Bal< 10k	1099.0 (1186.1)	1272.0 (990.3)	261.9 (4882.7)	-0.0004 (0.0197)
Opti× Post× Bal< 10k	-888.1 (1277.8)	-1981.6* (1140.5)	1093.9 (5604.6)	-0.0037 (0.0296)
Pre mean	10389.02	12863.14	88247.79	0.3153
N	944	944	944	944
N×T	7545	7545	7545	7545
Ind FE	X	X	X	X
Time FE	X	X	X	X

Notes: Table reports coefficients from estimating:

$$y_{i,t} = \beta_1 \text{Beliefs}^{pre} \times \text{Post}_t + \beta_2 \mathbb{1}\{\text{Balance}^{pre} < \$10k\} \times \text{Post}_t + \beta_3 \text{Beliefs}^{pre} \times \mathbb{1}\{\text{Balance}^{pre} < \$10k\} \times \text{Post}_t + \alpha_i + \alpha_t + \epsilon_{i,t}$$

Panel A. Estimation window from April 2022-December 2022, excluding August 2022 (partial treatment). Post_t refers to period from September 2022, the first full month following the 8/24/2022 forgiveness announcement. Beliefs^{pre} is an indicator for having below-median beliefs prior to the announcement, denoted *Pess* in the table. Panel B: Estimation window from March 2023-October 2023. Post_t refers to period from July 2023, following the June 30 Supreme Court decision, and Beliefs^{pre} is an indicator for having above-median beliefs prior to the Supreme Court decision, denoted *Opti* in the table. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Source: Credit and employment data

Table A.4: Credit and Employment Outcomes Around End of Payment Pause

	(1)	(2)	(3)	(4)
	Revolving Bal (\$)	Auto Bal (\$)	Mortgage Bal (\$)	Employment Any
Opti Repayment× Post	-446.7 (344.3)	-134.2 (455.0)	-5898.0** (2843.4)	-0.0151 (0.0115)
Opti Forgiveness× Post	-7.690 (345.9)	-490.7 (455.7)	2060.7 (2904.9)	0.0028 (0.0116)
Pre mean	11109.17	12424.29	95602.30	0.306
N	773	773	773	773
N×T	6956	6956	6956	6956
Ind FE	X	X	X	X
Time FE	X	X	X	X

Notes: Table reports coefficients from estimating:

$$y_{i,t} = \beta_1 \mathbb{1}\{\text{OptimistRepayment}^{Pre}\} \times \text{Post}_t + \beta_2 \mathbb{1}\{\text{OptimistForgiveness}^{Pre}\} \times \text{Post}_t + \alpha_i + \alpha_t + \epsilon_{i,t}$$

Estimation window from June 2023-February 2024. Post_t refers to the period from October 2023 when payments were again due. Sample is restricted to student loan borrowers with beliefs surveyed post the June 30 Supreme Court Decision. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Effect of Information Treatment on Credit and Employment Outcomes, Instrumental Variable Estimates

	(1)	(2)	(3)	(4)
	Revolving Bal (\$)	Auto Bal (\$)	Mortgage Bal (\$)	Employment Any
Prob Forgive $\times$ Post Treat	4877.8 (5582.1)	-6262.7 (6704.9)	-61072.0 (40200.6)	0.0453 (0.1563)
Post Treat	-1634.0 (1800.3)	2227.4 (2145.9)	19813.2 (13051.9)	-0.0106 (0.0482)
NxT	25791	25791	25791	25791
Individual FE	X	X	X	X
Year-Month FE	X	X	X	X
Pre Balance > 0	X	X	X	X
No Payment Required	X	X	X	X

Notes: Table reports results from an IV event-study framework:

$$y_{it} = \beta_{IV} \pi^1 \times \widehat{1\{Post\}} + \beta_2 1\{Post\} + \alpha_i + \alpha_t + \epsilon_{it}$$

$$\pi_{it}^1 \times 1\{Post\} = b_1 1\{OptiTreat\} \times 1\{Post\} + b_2 1\{Post\} + a_i + a_t + u_{it}$$

where $1\{Post\}$ is a post survey indicator, π^1 are one-year ahead beliefs about forgiveness, $1\{OptiTreat\}$ is an indicator for receiving the optimist information treatment, and α_i and α_t are individual and time fixed effects. The event window is 4 months around the survey month, excluding the survey month due to partial treatment. In all regressions, the control group received no information treatment or the pessimistic treatment. Some individuals are in multiple independent treatments in the event studies. Standard errors clustered at the individual level in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table A.6: Equivalent Variation Between Ex-Post Correct Belief and Selected Ex-Ante Beliefs

	(1)	(2)	(3)	(4)	(5)
	Fixed Within Repayment Plan			Optimal Repayment Plan	
	Forgiveness Realizations:			Forgiveness Realizations:	
	No Forgiveness	Forgiveness 3 Years from Origination	Forgiveness 10 Years from Origination	No Forgiveness	Forgiveness 10 Years from Origination
Choose 10-year repayment plan:					
0% Forgiveness	0.00	-0.08	0.00	0.00	-2.71
50th Percentile	0.00	-0.05	0.00	0.00	-2.71
Mean	-0.01	-0.04	0.00	-0.01	-2.71
100%	-0.08	0.00	0.00	-0.08	-2.71
Choose 20-year repayment plan:					
0% Forgiveness	0.00	-0.09	-0.11	-0.34	-0.11
50th Percentile	0.00	-0.06	-0.07	-0.35	-0.07
Mean	-0.01	-0.05	-0.06	-0.35	-0.06
100%	-0.09	0.00	0.00	-0.44	0.00

Notes: Table answers the question: “What constant percent of consumption would a borrower need to give up under ex-post correct beliefs, to make utility equivalent to the indicated beliefs?” Columns (1)-(3) compare differences to correct beliefs within a payment plan choice (10-year or 20-year). In columns (4)-(5), borrowers choose the repayment plan that maximizes their utility under their ex-ante beliefs about forgiveness. “No forgiveness” is relative to a counterfactual where forgiveness is expected after 3 years at the indicated beliefs. “Forgiveness” refers to receiving forgiveness on remaining balance. In our baseline calibration, borrowers are assumed to have CRRA preferences with CRRA parameter $\rho = 2$ and discount factor $\beta = 0.96$. The rate of borrowing for student loans, r_b , is set at 6.5 percent; the rate for saving and outside borrowing, r_s , is set at 5 percent. The first three years payments are assumed paused and the loan accrues no interest. Borrowers are otherwise assumed to be identical across beliefs and loan characteristics. Student loan balances, income and beliefs are calibrated based on credit bureau data and survey responses. Under this calibration, the 10-year repayment plan is optimal if no forgiveness is expected, and 20-year repayment plan is optimal if forgiveness is expected. Ex-post correct beliefs are in bold.

B Experimental Design

B.0.1 May-June 2022

Table B.1: Treatments, May-June 2022

Optimistic	Pessimistic
“President Biden signaled to House Democrats that he is seriously considering taking action to forgive student-loan debt on a large scale, according to congressional aides and others familiar with the discussions. Following a meeting with the president at the White House, members of the Congressional Hispanic Caucus said they felt confident that Mr. Biden is warming to calls from progressive Democrats to take executive action to erase at least some of the debt held by borrowers with federal student loans. The president told the lawmakers that he was weighing the timing of any announcement.” —WSJ, 4/28/2022	“President Biden would be on shaky legal ground if he were to pursue broad-based student debt cancellation by executive action, President Barack Obama’s former top Education Department lawyer wrote in a legal analysis. Using executive action to cancel debts for student borrowers without tying relief to their individual needs and using regulatory procedures would put the Biden administration at risk of having its plan overruled in court, according to a legal analysis prepared by Charlie Rose, who served as the top lawyer in the Education Department under President Obama from 2009 to 2011.” —WSJ, 5/4/2022

Sources: The optimistic article excerpted from *The Wall Street Journal* (Restuccia, 2022), and the pessimistic article is also excerpted from *The Wall Street Journal* (Rubin, 2022).

B.0.2 Nov-Dec 2022

Table B.2: Treatments, Nov-Dec 2022

Optimistic	Pessimistic
“Federal courts delivered two wins for President Joe Biden’s student loan forgiveness plan. Supreme Court Justice Amy Coney Barrett rejected a challenge to the program brought by a Wisconsin taxpayers group. And on the same day, a federal district court judge rejected a separate lawsuit brought by six Republican-led states. Student loan cancellations, worth up to \$20,000 per eligible borrower, could show up on borrowers’ accounts soon.” <i>Edited for brevity and readability.</i>	“A federal judge in Texas struck down the Biden administration’s student-debt forgiveness plan, imperiling a key administration priority that would have canceled up to \$20,000 in student loans for tens of millions of borrowers. The Biden administration’s plan is an “unconstitutional exercise of Congress’s legislative power” that also failed to go through normal regulatory processes, Judge Mark Pittman of the Northern District of Texas wrote in a 26-page opinion.” <i>Edited for brevity and readability.</i>

Sources: The optimistic article excerpted from *CNN* (de Vogue and Lobosco, 2022), and the pessimistic article is excerpted from *The Wall Street Journal* (Rubin and Restuccia, 2022).

B.0.3 April-June 2023

Table B.3: Treatments, April-June 2023

Optimistic	Pessimistic
“Is it a good idea for the government to cancel \$400 billion in student loans? That is not the question that the Supreme Court confronted, though you could be forgiven for thinking it was. Tuesday’s cases involved not the wisdom of loan relief but its legal merits—and, perhaps more importantly, the plaintiffs’ right to challenge the program in the first place. Defying predictions of doom for the Biden administration, the arguments went much better than expected... The Biden administration rooted the program in the HEROES Act, which Congress enacted in 2003 to replace a previous statute that granted loan relief to people affected by 9/11. It allowed the education secretary to “waive or modify” student debt held by borrowers affected by any national emergency—not just terrorism. And it expressly let the secretary forgive many, many loans in one fell swoop, rather than “case-by-case.” Initially set to expire in 2005, the HEROES Act was extended by Congress for another two years. Then, in 2007, Congress made the act permanent. By its plain text, the HEROES Act authorizes Biden’s plan.” <i>Edited for brevity and readability.</i>	“The Supreme Court’s conservative majority signaled deep skepticism about President Joe Biden’s plan to wipe out \$400 billion in student loan debt for tens of millions of Americans, suggesting the president overstepped his powers under the law. In one of a series of troubling signs for the plan, Chief Justice John Roberts questioned a central premise of Biden’s argument: that the president has authority to cancel student loans without explicit authority from Congress.” <i>Edited for brevity and readability.</i>

Sources: The optimistic article excerpted from *Slate* (Stern, 2023), and the pessimistic article is excerpted from *USA Today* (Fritze, Garrison, Quintana, and Wong, 2023).

Table B.4: Experiment Balance

	(1) Control	(2) Optimistic Treat	(3) Pessimistic Treat	(4) Difference (2)-(1)	(5) Difference (3)-(1)
Student Loan Outcomes					
Mean balance	68,517.71	71,850.39	72,837.92	3,332.68	4,320.21
Median balance	48,725.00	49,173.00	45,740.00	448.00	-2,985.00
Newest, age in years	6.26	6.27	6.28	0.01	0.02
Any deferred	0.32	0.33	0.33	0.01	0.01
Any delinquent	0.07	0.07	0.05	-0.00	-0.02
Any monthly payment	0.46	0.48	0.45	0.02	-0.01
Mean monthly payment if >0	389.56	382.48	391.19	-7.08	1.63
Other Credit and Employment Outcomes					
Any revolving balance	0.83	0.83	0.83	-0.00	-0.01
Mean revolving balance if >0	10,784.00	11,084.57	10,734.16	300.57	-49.84
Any auto balance	0.60	0.62	0.60	0.02	-0.00
Mean auto balance if >0	20,163.92	18,842.81	18,944.21	-1,321.11	-1,219.70
Any mortgage balance	0.43	0.43	0.42	-0.00	-0.01
Mean mortgage balance if >0	187,791.61	188,864.95	179,787.47	1,073.34	-8,004.14
Vantage 4.0 Credit Score	660.95	659.75	663.83	-1.21	2.87
Borrower Characteristics					
Age	46.14	46.20	46.16	0.06	0.02
Female	0.58	0.58	0.58	0.00	-0.00
Married	0.47	0.47	0.50	0.00	0.03
Has children	0.48	0.47	0.50	-0.01	0.02
Zipcode and County Characteristics					
Zipcode AGI (2019)	65,856.78	66,417.83	66,412.26	561.04	555.48
Zipcode ZHVI Home prices (Feb 2020)	245,608.34	251,427.05	250,235.12	5,818.70	4,626.78
Zipcode $\Delta \ln(\text{ZHVI})$, Dec 2023-Feb 2020	0.34	0.33	0.33	-0.00	-0.01*
County COVID mortality cumul. per 100k	319.57	310.28	318.33	-9.29**	-1.24
Observations	963	967	964		

Notes: Table shows summary statistics comparing those in the control group with the optimistic and pessimistic treatment groups. Outcomes shown for February 2020 (pre-COVID payment pause) unless otherwise specified. Stars denote statistical significance of differences in means using robust standard errors, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Source: Zipcode AGI from IRS Tax Stats; ZHVI from Zillow Research; COVID mortality from Opportunity Insights. Other outcomes from credit and employment data.

C Additional Survey Results

Figure C.1: Beliefs about the Supreme Court Decision (Surveyed April-June 2023)

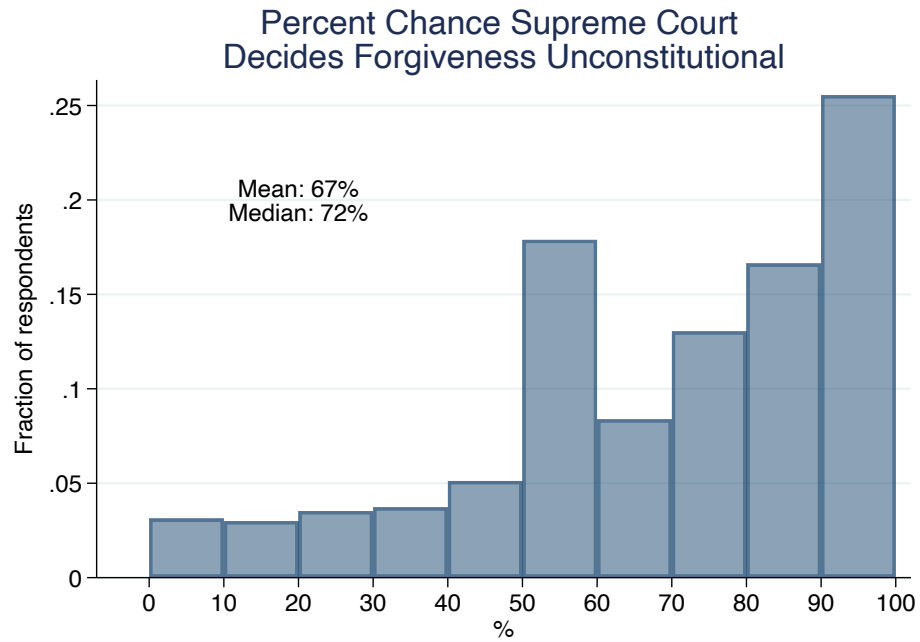


Figure C.2: Expected Sources of Forgiveness (Post SCOTUS Decision)

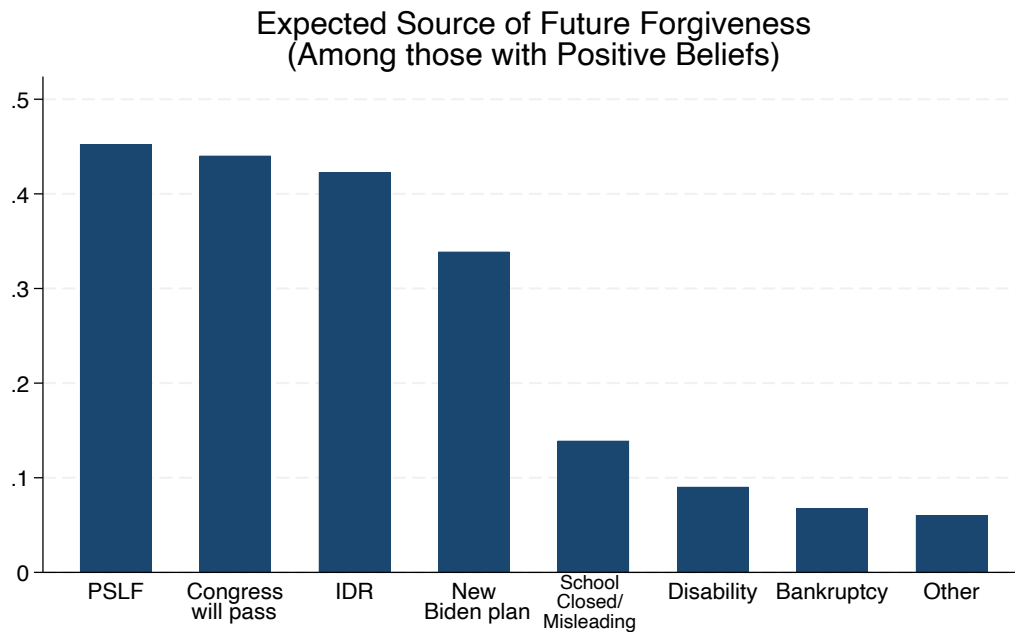
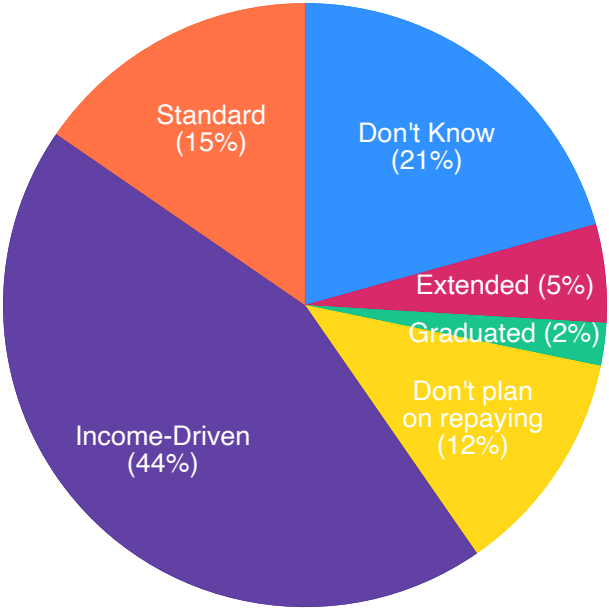


Figure C.3: Intended Plan Enrollment at Repayment



D Survey Instrument

D.1 Main Survey, June-Aug 2022

Thank you very much for your interest in our survey, which involves academic research. The purpose of the research is to better understand the pros and cons of several proposed reforms regarding debt forgiveness. There is no right or wrong answer. Answers are only right if they truly reflect your own views and opinions, as opposed to those of somebody else.

We do not foresee any type of risk or discomfort from this survey. Participation is entirely voluntary, and you will be able to leave the survey at any time you want if you faced unknown risks. Withdrawal from participation will not hinder you from learning about the economic reforms and provide your opinions about them.

For the analysis, the researchers need to merge your record with credit bureau data. Your answers will be analyzed in aggregation with the answers of all other respondents. All your answers will be confidential. The researchers who will evaluate the data will have no access to any identifiable information about you.

If you agree with these conditions, and are willing to participate in the survey, please check the box below:

I consent to the conditions to participate in the survey.

☐ Yes

☐ No

Have you ever taken out a student loan to help pay for your own education? This would exclude loans you took out for any other person (e.g. ParentPLUS loans).

☐ Yes

☐ No

☐ Don't know

What is your approximate amount of all your student loans, both federal and/or private student loans, for both your undergraduate and/or graduate education, that you owe for your own education as of today?

☐ Student loan balance (\$) _____

☐ Don't know

If “Don’t know”:

Which of the following categories best describes the approximate amount of all your student loans, both federal and/or private student loans, for both your undergraduate and/or graduate education, that you owe for your own education as of today?

- ☐ None (\$0)
 - ☐ \$1-4,999
 - ☐ \$5,000-9,999
 - ☐ \$10,000-14,999
 - ☐ \$15,000-19,999
 - ☐ \$20,000-29,999
 - ☐ \$30,000-39,999
 - ☐ \$40,000-49,999
 - ☐ \$50,000-99,999
 - ☐ \$100,000 or more
 - ☐ Don’t know
-

In February 2020, prior to the start of the COVID-19 pandemic, do you remember what kind of repayment plan you were enrolled in?

- ☐ None, I was still in school
- ☐ Standard Repayment Plan: Make the same payment amount every month for 10 years
- ☐ Graduated Repayment Plan: Payment amounts start low and increase every two years. Loans are paid off within 10 years
- ☐ Extended Repayment Plan: Make a payment amount every month for longer than 10 years (up to 30 years)
- ☐ Income-Driven Repayment Plan (IDR): Monthly payment amounts are tied to your income and family size
- ☐ None, Loans were in default
- ☐ Don't know

In March 2020, the federal government announced that federal student loan payments and interest would be temporarily paused for most borrowers as part of the pandemic assistance efforts and this “payment pause” is still in place.

Prior to this taking this survey, were you aware that the federal government paused payment for most federal student loans?

- ☐ Yes
- ☐ No

During the payment pause related to the COVID-19 pandemic, have you continued to make payments on your federal student loan(s)?

- ☐ Yes, I am still required to make monthly payments
 - ☐ Yes, I voluntarily make full payments each month
 - ☐ Yes, I voluntarily make partial or full payments when I can, but not every month
 - ☐ No, I haven't made any payments since payments were paused
 - ☐ Don't know
-

What kind of repayment plan are you currently enrolled in?

- ☐ Standard Repayment Plan: Make the same payment amount every month for 10 years
 - ☐ Graduated Repayment Plan: Payment amounts start low and increase every two years. Loans are paid off within 10 years
 - ☐ Extended Repayment Plan: Make a payment amount every month for longer than 10 years (up to 30 years)
 - ☐ Income-Driven Repayment Plan (IDR): Monthly payment amounts are tied to your income and family size
 - ☐ Don't know
-

The current payment pause is expected to end in August 2022.

To what extent do you agree with the following statement: When student loan payments start again in September 2022, I will be able to afford the same monthly amount that I was paying prior to the payment pause related to the COVID-19 pandemic.

- ☐ Strongly Agree
 - ☐ Agree
 - ☐ Neither agree nor disagree
 - ☐ Disagree
 - ☐ Strongly disagree
-

On the next page, you will be shown a short article that features a statement about the likelihood of student loan forgiveness. Then you will be asked some questions about the article you were shown.

If Treatment 1:

President Biden signaled to House Democrats that he is **seriously considering taking action to forgive student-loan debt on a large scale**, according to congressional aides and others familiar with the discussions.

Following a meeting with the president at the White House, members of the Congressional Hispanic Caucus said they felt confident that Mr. Biden is warming to calls from progressive Democrats to take executive action to erase at least some of the debt held by borrowers with federal student loans. The president told the lawmakers that he was weighing the timing of any announcement.

Edited for brevity and readability

If Treatment 2:

President Biden would be on **shaky legal ground if he were to pursue broad-based student debt cancellation** by executive action, President Barack Obama's former top Education Department lawyer wrote in a legal analysis.

Using executive action to cancel debts for student borrowers without tying relief to their individual needs and using regulatory procedures would put the Biden administration at risk of having its plan overruled in court, according to a legal analysis prepared by Charlie Rose, who served as the top lawyer in the Education Department under President Obama from 2009 to 2011.

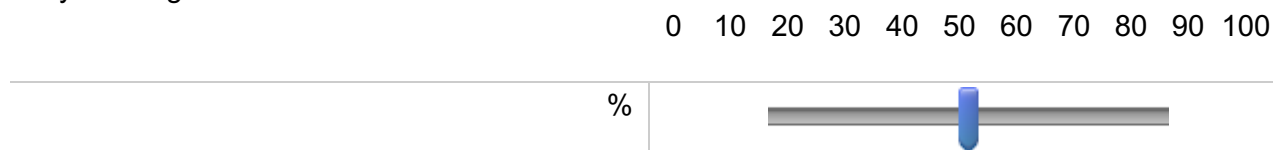
Edited for brevity and readability

In the following questions, we will ask you to think about the percent chance of something happening in the future.

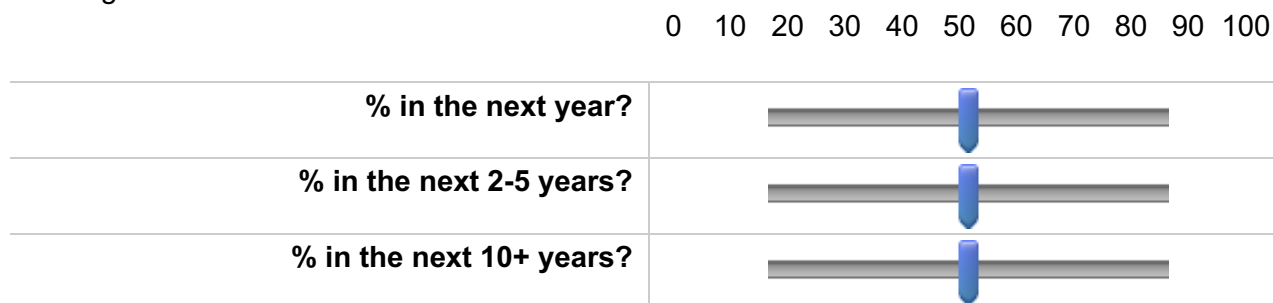
Your answers can range from 0 to 100, where 0 means there is absolutely no chance, and 100 means that it is absolutely certain.

For example, numbers like: 2 and 5 percent may indicate "almost no chance" 18 percent or so may mean "not much chance" 47 or 52 percent chance may be a "pretty even chance" 83 percent or so may mean a "very good chance" 95 or 98 percent chance may be "almost certain"

What do you think is the percent chance that the repayment pause will be extended again beyond August 2022?

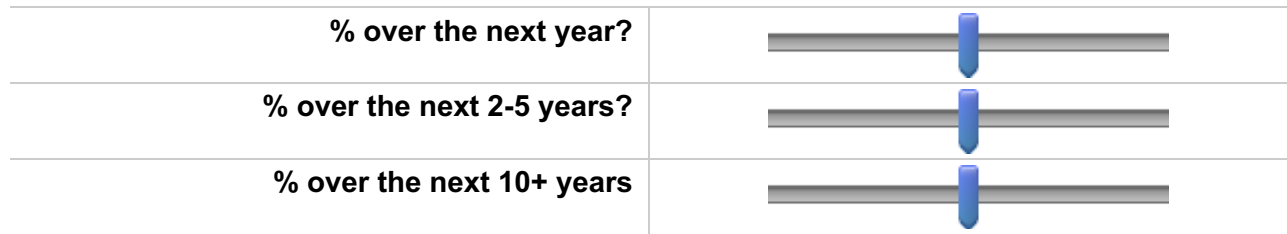


What do you think is the percent chance that **AT LEAST SOME of your** student loan debt will be forgiven.



Which fraction of your student loan do you think will be forgiven?

0 10 20 30 40 50 60 70 80 90 100



If student loan forgiveness is announced, what dollar value of student loans do you think will be forgiven?

☐ Dollars (\$) _____

☐ Don't know

Approximately how much savings do you have available in your bank accounts?

- ☐ None (\$0)
- ☐ \$1-4,999
- ☐ \$5,000-9,999
- ☐ \$10,000-14,999
- ☐ \$15,000-19,999
- ☐ \$20,000-29,999
- ☐ \$30,000-39,999
- ☐ \$40,000-49,999
- ☐ \$50,000-99,999
- ☐ \$100,000 or more
- ☐ Don't know

How likely are you to vote in the midterm elections in November 2022?

Not at all

For sure

0 1 2 3 4 5 6 7 8 9 10

1



In the Congressional elections in November 2022, which party are you likely to vote for?

- ☐ Democrat
 - ☐ Republican
 - ☐ Not sure
-

D.2 Survey Intentions, April-July 2023

Display This Question:

If If What is your approximate amount of all your student loans, both federal and/or private student lo... Text Response Is Greater Than or Equal to 10000

Suppose you received student loan forgiveness of \$10,000 in July 2023. How many dollars would you allocate to each of the following this year?

To pay off debts: : _____ (1)
To new spending: A house/ apartment : _____ (2)
A car or other vehicle : _____ (6)
A large home appliance or electronics : _____ (5)
All other spending: : _____ (4)
Amount saved: : _____ (3)
Total : _____

Display This Question:

If If What is your approximate amount of all your student loans, both federal and/or private student lo... Text Response Is Less Than 10000

Suppose your student loans are completely forgiven in July 2023. How many dollars would you allocate to each of the following this year?

Pay off debts: : _____ (1)
New spending this year: A house/apartment : _____ (2)
A car or other vehicle : _____ (3)
A large home appliance or electronics : _____ (4)
All other spending: : _____ (5)
Amount saved: : _____ (6)
Total : _____

Are you currently employed?

- ☐ Yes, full time (1)
- ☐ Yes, part-time, but want full-time (2)
- ☐ Yes, part-time by choice (3)
- ☐ Student and not employed, or employed part-time (4)
- ☐ No, unemployed and looking for work (5)
- ☐ No, not employed, not a student, and not looking for work (6)

Display This Question:

If Are you currently employed? = Yes, full time

Or Are you currently employed? = Yes, part-time, but want full-time

Or Are you currently employed? = Yes, part-time by choice

Or Are you currently employed? = Student and not employed, or employed part-time

If up to \$10,000 in your student loans are forgiven in July 2023, would you change work hours in your current job(s) or possibly a new job?

- ☐ No, I would not change my hours or job (1)
- ☐ I will try to increase my work hours in my current job (2)
- ☐ I will try to decrease my work hours in my current job (3)
- ☐ I will look for a new job more actively (4)
- ☐ I will look for a new job with about the same effort (5)
- ☐ I will look for a new job less actively (6)

Display This Question:

If Are you currently employed? = No, unemployed and looking for work

Or Are you currently employed? = No, unemployed and looking for work

If up to \$10,000 in your student loans are forgiven in July 2023, would you change your effort in looking for a job?

- ☐ I will look for a job more actively (1)
- ☐ I will look for a job with about the same effort (2)
- ☐ I will look for a job less actively (3)